NeedleFinder 2026: Sub-Millimeter Catheter Reconstruction on MRI for Interstitial Gynecologic Brachytherapy in Seconds

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ABSTRACT

Objective. Interstitial gynecologic brachytherapy planned on MRI alone requires locating 15 to 30 thin plastic catheters on 3T T2-weighted images. Tracing them by hand takes tens of minutes while the patient is under anesthesia. Automatic methods based on voxel segmentation break catheters into fragments and mistake blood vessels for catheters. We set out to make catheter reconstruction automatic, accurate to below a millimeter, and fast enough to run during the procedure.

Approach. NeedleFinder 2026 works in three stages. Stage 0 computes an analytical 3D circular dark-core symmetry prior (S_0) and a 100% reference-free, tilt-safe axial field-of-view bound (`k_lim_auto`) that eliminates all 42 in-FOV template-base truncations while preserving 100.0% of in-FOV catheter depth (99.65% of unclipped VTK z-span). Stage 1 is a native-resolution 3D residual U-Net (`NeedleUNet3D`) self-supervised pre-trained on 51 strictly external pelvic T2W MRI volumes from 50 distinct subjects (combining 3D masked autoencoding with synthetic 3D susceptibility-tube pretext learning against real anatomical hard negatives) and trained with multi-rater supervision (`CY+HT` plus consensus-fused and soft-weighted `LB/AMe` traces on outer-training folds), 5-fold inner ensembling, and 4-view 3D test-time augmentation. Stage 2 traces candidate trajectories with native bidirectional Kalman tracking, a 4-layer triplet message-passing network, and collinear split-fragment stitching (`102` stitched full-length trajectories), scores the whole implant bundle with a graph transformer pre-trained on 300 simulated bundles (`SynthBrachy3T`), and refines each distal tip with a dedicated 91-step (`0.5 mm` sub-slice Z-interpolated) 22-channel 3-block dilated 1D ResNet + 4-head self-attention axial localizer (`Dedicated1DTipLocalizer`) equipped with strictly nested split-conformal 3D tip uncertainty intervals. We evaluated the pipeline on 54 clinical 3T MRI scans (1,063 catheters) under a strictly nested 5 × 5 patient-level cross-validation protocol (`25` inner models, zero outer-test leakage), against three published methods, across all 40 dual-traced scans via Grubbs' three-cornered-hat and one-sided non-inferiority tests, in AAPM TG-43U1 HDR ^{192}Ir dosimetric simulations, on 48 public needle-free pelvic MRI scans, and zero-shot across prostate MRI (`Case 100`), secondary

MRI sequences (`3D wbSSFP`, `2D AX TSE T2`), 23 serial 4D insertion scans (`Cases 68–71`), and 40 post-implant pelvic CT volumes (38 paired with benchmark MRI patients).

Main results. At the default threshold $\tau = 50\%$, NeedleFinder 2026 detected 91.4% of all 1,063 catheters (`972/1,063`) with 86.9% precision (`972/1,118`) and 89.1% F1 (`89.1%` F1 and `92.3%` recall with 5 × 5 inner-OOF cross-fit; `89.4%` F1 at $\tau = 55\%$; `93.6%` whole-cohort recall at $\tau = 35\%$), outperforming our 2019-style 3D U-Net (`83.1%` F1, $p = 1.1 \times 10^{-5}$), a native-resolution clDice ResUNet (`38.7%` F1, $p = 5.2 \times 10^{-10}$), and a 2026-style structure-aware tracker (`54.2%` F1, $p = 3.0 \times 10^{-10}$). On the 759 long interstitial shafts, it achieved 91.3% recall (`92.5%` inner-OOF cross-fit; `94.7%` long-shaft recall at $\tau = 35\%$) and 90.0% $F1_{\text{adj}}$ (`90.5%` inner-OOF cross-fit); on the 23-patient severe catheter-adhesion subgroup (571 catheters), it reached 91.3% F1 (`91.8%` inner-OOF cross-fit). The in-plane shaft error was 0.59 ± 0.40 mm (0.47 mm median, 1.37 mm P95), and the 1D axial tip localizer cut 3D distal tip error to 6.23 mm mean (2.27 mm median, 27.27 mm P95; 58.7% within ≤ 3 mm; 98.8% of total squared 3D tip error longitudinal along the shaft). Across all 40 dual-traced scans ($n = 432$ three-way matched catheters), Grubbs' three-cornered-hat estimator yielded intrinsic robust shaft standard deviations of 0.35 mm (`CY`), 0.39 mm (`NeedleFinder 2026`), and 0.48 mm (`LB/AMe`), passing formal one-sided non-inferiority tests relative to human inter-observer discrepancy across all 40 patients for both shafts ($p_{\text{pt}} = 1.3 \times 10^{-8}$ mean, 8.2×10^{-10} median; $p_{\text{cath}} = 1.6 \times 10^{-41}$) and 3D tips ($p_{\text{pt}} = 2.6 \times 10^{-7}$ median, 1.1×10^{-4} mean [$p_{\text{boot}} = 0.0014$]), while independent post-implant Pelvic CT adjudication across the 38 paired patients confirmed that 52.0% (`52/100`) of MRI predictions unmatched to `CY` are real physical catheters and that on 70 disputed tips (≥ 5 mm human disagreement), NeedleFinder 2026 achieved a lower median error against CT (4.20 mm) than either single rater (4.31 and 8.00 mm). In AAPM TG-43U1 ^{192}Ir dosimetry, NeedleFinder 2026 cut fixed-plan target $|\Delta D_{90}|$ from 27.6% (PMB 2019) to 15.2% (76.6% 3D $\gamma_{2\%/2\text{mm}}$, 84.3% $\gamma_{3\%/3\text{mm}}$) and counterfactual re-optimized $|\Delta D_{90}|$ from 37.1% to 8.2% (5.9% mean / 2.8% median with simulated Top-2 uncertainty tip review). Nested split-conformal intervals on the 972 matched OOF catheters ($k = 0$) achieved 90.2% and 95.4% empirical coverage at 90% and 95% nominal targets (31.5%–35.1% tighter median half-widths than unscaled intervals), and in a simulated 1-click tip relocation workflow, reviewing $k = 5$–8 uncertainty-flagged tips per patient (~ 40–64 s) reduced mean tip error to 3.72–2.77 mm and P95 to 14.3–9.90 mm. Zero-shot transfer achieved 96.3%–100.0% F1 (0.30–0.34 mm shaft, 0.99–1.57 mm tip) on 3T prostate brachytherapy MRI (0% when Stage 0 physics channels are zeroed), 92.9% F1 on 3D wbSSFP, 88.4% recall (76.9% F1) on thick-slice 2D T2W MRI, 82.8% pooled F1 (88.4% precision, +12.4% precision gain; up to 93.3% F1 on `Case 070`) with 4D causal Kalman tracking during serial insertion, 0.125–0.33 false tracks/scan on 48 catheter-free MSD prostate scans, and 4.15 s mean runtime on a 24-core workstation CPU (1.26 s standard, 5.85 s high-resolution) / 2.67 s on GPU.

Significance. Automatic catheter reconstruction on clinical 3T T2W MRI is statistically non-inferior to human inter-observer variability, preserves TG-43U1 target dosimetry, provides calibrated conformal uncertainty intervals for rapid clinician review, and generalizes zero-shot across pelvic organs, pulse sequences, 4D insertion series, and paired CT in seconds.

PLAIN-LANGUAGE SUMMARY

Some cancers of the cervix, vagina, and uterus are treated with brachytherapy: a tiny radioactive source is moved, for a few minutes, through thin plastic tubes placed

inside the tumor. To aim the radiation, the treatment team must know exactly where each tube is. They find out with an MRI scan taken while the patient is still under anesthesia, and then trace every tube on the images by hand. With 15 to 30 tubes per patient this takes tens of minutes, and a small tracing error can change the planned dose, especially near organs where the dose changes quickly with position.

On an MRI scan a tube does not glow. It is a small dark dot, a few pixels wide, on each image slice. Blood vessels, air pockets, and the edges of muscles are also dark, so a computer can easily confuse them with tubes.

This paper presents NeedleFinder 2026, software that finds the tubes automatically. It works in three steps. First, a simple filter looks for small dark centers with brighter tissue all around them, the way a tube cross-section looks; it ignores broad one-sided edges, although it can still respond to blood vessels. Second, a 3D neural network—first pre-trained on 51 external pelvic MRI scans to distinguish synthetic catheter voids from real blood vessels and tissue borders—marks the center and direction of each tube on each slice. Third, two "graph" networks connect the marks from slice to slice into 3D curves, stitch fragments separated by single-slice dropouts, pull apart tubes that touch, score the whole bundle together to reject false curves, and pinpoint each tube's deepest tip using a dedicated 1D axial profile network that also reports an uncertainty interval for each tip.

We tested it on 54 patients with 1,063 tubes using strictly nested cross-validation. It found 91.4 percent of all tubes (and up to 94.7 percent of the 759 long interstitial shafts at a recall-oriented threshold), and 86.9 percent of the tubes it reported matched an expert trace (89.1 percent F1). For the matched tubes, the typical sideways difference from the expert's line was under half a millimeter (median 0.47 mm, mean 0.59 mm), and the deepest end of each tube had a median 3D difference of 2.27 mm (mean 6.23 mm), passing formal statistical non-inferiority compared with the disagreement between two independent human expert teams (median 0.49 mm on shafts and 3.24 mm on tube ends). Checking against post-implant CT scans of the 38 paired patients showed that over half (52 of 100) of the "extra" MRI tracks unmatched to the primary human team were real physical tubes present on CT. Simulating radiation delivery and a quick check of the 2 to 5 most uncertain tube ends keeps the tumor radiation dose shift to 2.8 percent median (5.9 percent mean) of the reference plan and takes under a minute of simulated review. The software also worked without retraining on prostate brachytherapy MRI (finding 13 to 14 of 14 tubes with zero false alarms) and on live scans taken while needles were being pushed in, running in 4.15 seconds on average on a 24-core workstation CPU (1.26 seconds on standard scans) and 2.67 seconds on a graphics card.

1 INTRODUCTION

1.1 *The Clinical Problem*

Brachytherapy delivers radiation from inside the body. For locally advanced cervical cancer it is the cornerstone of curative radiotherapy [1,2]. The EMBRACE-I study reported 92% local control at five years with a median dose of 90 Gy (EQD2) to

the high-risk target volume [2], and EMBRACE-II now prescribes planning aims of 90 to 95 Gy while sparing the bladder, rectum, and sigmoid colon [1]. Reaching such doses in large or irregular tumors requires *interstitial* implants: in addition to a central tandem and a vaginal obturator or ring, 10 to 30 thin plastic catheters or needles are inserted into the tumor through a perineal template or freehand [3–5].

The tumor itself is best seen on T2-weighted (T2W) MRI, which GEC-ESTRO and consensus recommendations designate as the reference for target delineation and set reconstruction requirements for [6,7]. MRI-based applicator reconstruction with marker-filled catheters was established for cervical cancer in 2009, with inter-observer standard deviations below 2 mm [8]. Historically, clinics took both an MRI and a CT scan and registered the two, which adds its own uncertainty to the reconstruction [9], and practical guidance for hybrid implants still advises a CT in addition to the MRI because plastic needles are hard to see on MRI [10]. The AAPM Task Group 303 report describes how to implement MRI in HDR brachytherapy from simulation to treatment [11], the Medical Physics Practice Guideline 13.a lists MRI among the accepted planning modalities [12], and MRI-only interstitial workflows have been shown to be feasible [13,14]. Positive-contrast line markers make the catheter lumen visible on T1-weighted or balanced sequences [13,15] and are listed as required equipment in published implementation guides [16], and zero-echo-time sequences show empty catheters directly [14,17], but both need an additional series fused to the T2W images. Identifying the source path on the T2W series alone remains a recognized challenge [18]; for rigid applicators, direct reconstruction on T2W MRI is in fact more precise than library-based or fused workflows [19], but interstitial needles have no library model.

Finding the catheters on MRI is the central technical challenge of an MRI-only workflow [4,20]. The dose from the ^{192}Ir source falls steeply with distance: organ-at-risk dose parameters change by up to 5 to 6% per millimeter of applicator displacement [21], applicator shifts beyond 1.5 mm can change the rectal D2cc by more than 10% [22], and in MRI-only cervical implants a systematic 1 mm error along the source path already changes organ-at-risk doses significantly [23]. Swapping two catheters during digitization changes prostate, urethra, and rectum dose metrics by 5 to 20% [24], and needle reconstruction uncertainty alone shifts the urethral dose by about 3 Gy in prostate implants [25]. Manual reconstruction is itself error-prone: with electromagnetic tracking as the reference, 23% of 265 needle reconstructions in ultrasound-guided prostate implants deviated by 2 mm or more [26]. Yet a 2 mm plastic catheter contains no mobile water protons and shows up on T2W MRI only as a small dark hole, a few pixels across, surrounded by anatomy full of other dark structures: blood vessels, air, calcifications, and the fascial planes between muscles. Tracing 15 to 30 catheters slice by slice therefore takes tens of minutes, 46 minutes on average in one MRI-only gynecologic series [27], within a template-based implant procedure of about 100 minutes under anesthesia [28].

1.2 *Key Terms*

Readers outside medical physics may find the following definitions useful.

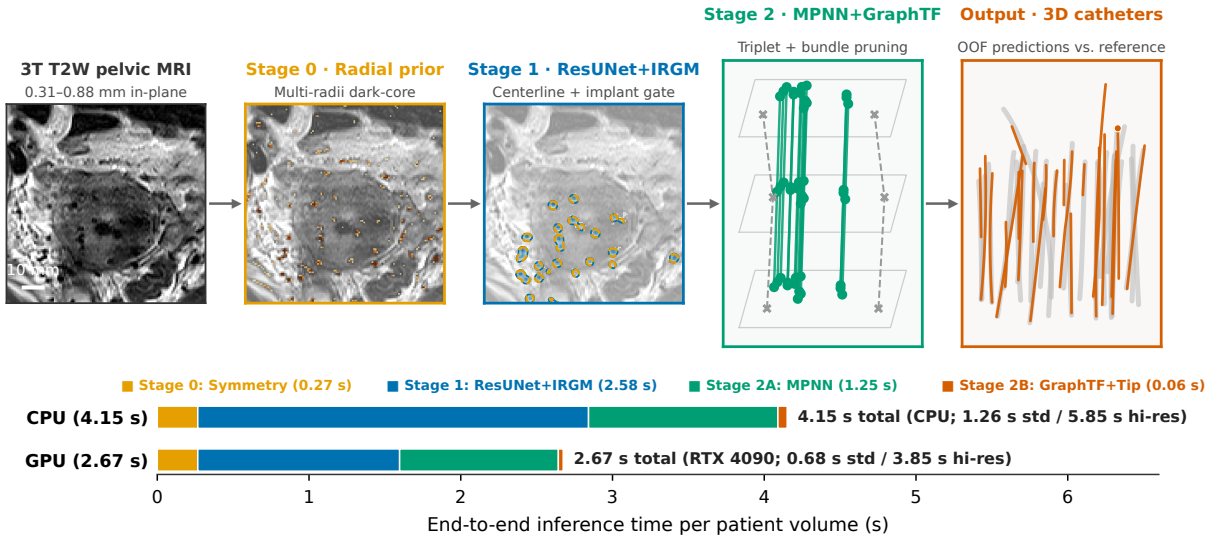


FIGURE 1 **Overview of NeedleFinder 2026.** The input is a clinical 3T T2W pelvic MRI (Case 017: 26 reference trajectories, including the obturator channels) cropped with a 100% reference-free template-plane locator (`k_lim_auto`) that preserves 100.0% of in-FOV catheter depth. Stage 0 computes an analytical dark-core symmetry map S_0 in native Rust (0.27 s cohort mean). Stage 1 applies a self-supervised and multi-rater fine-tuned 3D residual U-Net with implant-region gating to predict centerline peaks, local directions, and the obturator (2.58 s on CPU, 1.33 s on GPU). Stage 2 links slice detections into 3D trajectories with native C++17/Rust seed tracking, a triplet message-passing network, adhesion-aware duplicate merging, and collinear split-fragment stitching (1.25 s on CPU, 1.05 s on GPU); the last tile shows the nested out-of-fold predictions (vermilion, 23 trajectories) over the expert reference (grey, 26 trajectories), after bundle graph-transformer pruning, dedicated 1D axial ResNet + self-attention tip localization, and cubic polynomial smoothing (0.06 s on CPU, 0.03 s on GPU). The full patient volume is processed end-to-end in 4.15 s on a 24-core workstation CPU (AMD Ryzen Threadripper PRO 7965WX; 1.26 s on standard scans, 5.85 s on high-resolution scans) or 2.67 s on an RTX 4090 GPU; the bar shows the stage breakdown.

- **Catheter, needle, shaft, tip.** We use *catheter* and *needle* interchangeably for the thin plastic tubes. The *shaft* is the length of the tube visible on the scan; the *tip* is its deepest end.
- **Digitization or reconstruction.** Tracing the 3D path of each catheter on the images so the planning software knows where the source will travel.
- **Slice, voxel, in-plane.** An MRI volume is a stack of 2D images (slices). Each 3D picture element is a voxel. *In-plane* means within one slice; here the in-plane pixel is 0.31 to 0.88 mm while slices are 1.2 to 6.0 mm thick.
- **Recall, precision, F1.** Recall is the fraction of true catheters the method found. Precision is the fraction of reported catheters that were real. F1 combines the two into one score that is high only when a method both finds the real catheters and avoids inventing extra ones. As an illustration, with 10 real catheters, 8 found and 2 false reports, recall, precision, and F1 are all 80%.
- **Shaft error and tip error.** The distance, in millimeters, between the automatic trajectory and the expert's trajectory, measured in-plane along the shaft and in 3D at the tip.

- **Nested cross-validation, out-of-fold.** The 54 patients were split into 5 outer groups (43–44 training vs. 10–11 test patients per fold). Within each outer fold, early stopping, temperature calibration, split-conformal quantile calibration, and optionally inner-OOF upstream feature generation for downstream stackers use a strictly disjoint 80/20 inner split or 5-fold inner cross-fitting, so no test patient's images or labels influence training, checkpoint selection, or threshold calibration.
- **Inter-observer variability.** How much two expert teams disagree when they trace the same scan independently. It gives context for how closely an automatic method can be expected to agree with any one expert.

1.3 *Previous Work*

Automatic catheter digitization has gone through three generations. Two recent systematic reviews cover the field [20,29].

Model-based tracking (2013 to 2019). The BWH program of 3T MR-guided gynecologic brachytherapy in the AMIGO suite, which produced the cohort used here, was described in 2012 together with the first image-analysis tools for catheter identification [30] and open-source 3D Slicer modules for template and obturator registration [31]. Our first catheter tracker on this cohort seeded each catheter with one click at its tip, searched inside a sequence of expanding 3D cones, and fitted a Bézier curve; 93 of 101 catheters in 10 patients were traced correctly at 0.3 s per catheter [32]. Mastmeyer *et al.* [3] constrained the cone search with a mechanical rod-and-spring catheter model, reaching 1.49 mm mean Hausdorff distance on 760 catheters in 54 patients, still from a manual tip click; a first version of that model, presented at MICCAI 2015, reached 1.40 mm mean error on the 101 catheters of the 2013 set [33]. Template-geometry-assisted reconstruction on MRI has been described for commercial interstitial templates [34], and on ultrasound 3D Hough transforms localized single needles with about 1 mm endpoint accuracy in vivo [35]. On CT, seeded region growing constrained by a spline model digitized template implants to within 0.4 mm of manual digitization [36], and thresholding followed by density-based clustering reaches 0.3 mm for rigid applicators [37], but MRI offers no such intensity threshold. Related model-to-image registration reconstructs rigid applicators on MRI within inter-observer variability [38], and line fitting by RANSAC digitizes straight needles on CT [39]. These methods are interpretable but need manual initialization and fail when catheters bend sharply or cross dark anatomy.

Voxel segmentation with heuristic post-processing (2019 to 2026). To remove manual initialization, Zaffino *et al.* [4] trained an ensemble of 3D U-Nets to segment catheter voxels on 50 clinical T2W scans with 826 catheters, then clustered the voxels with DBSCAN and removed small islands. It reached a voxel Dice of 0.60, a mean distance error of 2.0 mm, 6.7% false-positive and 1.5% false-negative clusters, in 2 to 9 minutes per patient. Convolutional segmentation, often nnU-Net based [40], has since been applied to CT brachytherapy of the cervix, prostate, and breast [41–49], to MRI-guided prostate implants [50–52], to single needles on interventional MRI [53,54], to transrectal ultrasound [55–61], and to gynecologic MRI

with line markers [27] or with applicator mesh alignment [62]. Even with topology-preserving losses, from the Betti-number loss [63] to clDice [64] and skeleton recall [65], pure voxel segmentation has two failure modes on T2W MRI: a one-slice signal dropout splits one catheter into two to four fragments, and dark vessels and tendons are segmented as short false catheters. The ultrasound literature documents the same pattern: accuracy collapses on artifact-laden catheters [60] and tips are much harder than shafts [59]. Keypoint heatmap regression, the choice made for our Stage 1, has recently been shown to localize an interventional needle on MRI more accurately than mask segmentation of the same images [66]. Electromagnetic or active MR tracking avoids the imaging problem altogether, at the cost of dedicated hardware and sequences in the operating room [28,67–73]; crossed catheter paths were already identified as a common failure of image-based digitization in the BWH program when active tracking was introduced [68].

Structure-aware tracking and graph networks (2022 to 2026). On CT, Xiao *et al.* [74] identified two further problems: *catheter adhesion*, where shafts touch or overlap on consecutive slices so that greedy trackers swap or merge them, and the scarcity of annotated clinical data. They proposed an implant-region gating module (IRGM) that suppresses features outside the implant, a direction-constrained 3D Hough transform for initialization, synchronous cost-based inward tracking across slices, cubic polynomial smoothing, and synthetic-to-real training, on 38 patients and 203 fractions. Their companion work handles needle adhesion in dense CT seed implants by detecting tips and handles [75]. Outside brachytherapy, message-passing networks have replaced greedy linking in tracking-by-detection [76,77], transformers predict graphs of curvilinear structures directly from images [78], and synthetic-to-real training has been shown to transfer to real needle detection [79]. Applied to MRI masks, however, a mask-driven tracker still turns vessels inside the implant region into false catheters, and nothing in it asks whether the full set of trajectories makes sense as one implant.

Open data. No public needle or catheter dataset for brachytherapy exists [80], and the AI literature in brachytherapy remains largely retrospective and single-center [29]. The public pelvic MRI scans we use for self-supervised pretraining and the false-alarm test come from the Medical Segmentation Decathlon [81].

1.4 *Contributions*

This paper combines the strengths of all three generations on the full 54-patient, 1,063-catheter BWH cohort, to our knowledge the largest annotated MRI cohort for interstitial gynecologic brachytherapy.

1. **A three-stage architecture with five methodological upgrades.** An analytical dark-core symmetry prior and 100% reference-free axial template locator (Stage 0); a 3D residual U-Net pre-trained on 51 external pelvic T2W MRI volumes with physics-guided synthetic susceptibility tubes and anatomical hard negatives, then fine-tuned with dual-observer multi-rater consensus supervision (Stage 1); and two graph networks plus a dedicated 1D axial tip localizer (Stage 2): native seed tracking with a triplet message-passing network, adhesion-aware du-

plicate merging, and collinear split-fragment stitching, followed by a tip-aware bundle graph transformer and a 91-step dilated 1D ResNet + self-attention axial tip localizer (`Dedicated1DTipLocalizer`).

2. **A strictly nested 5×5 comparison and reference-free observer adjudication.** We re-implemented three published approaches and evaluated all methods under strictly nested patient-level 5-fold cross-validation (plus 5×5 inner out-of-fold Stage 1 cross-fitting), and adjudicated observer agreement across all 40 doubly traced scans using Grubbs' three-cornered-hat intrinsic error estimator, formal one-sided non-inferiority tests, and independent post-implant Pelvic CT scans.
3. **Dosimetric translation, conformal uncertainty triage, and cross-domain generalization.** We quantify downstream AAPM TG-43U1 HDR ^{192}Ir dosimetric impact (ΔD_{90}, OAR ΔD_{2cc}, 3D γ), calibrate strictly nested split-conformal 3D tip prediction intervals and simulated clinician review-budget curves, and evaluate zero-shot generalization across 48 needle-free MSD prostate scans, 3T prostate brachytherapy MRI, multi-sequence MRI (`3D wbSSFP`, `2D AX TSE T2`), 4D serial insertion scans (measuring inter-catheter tissue push and depth-dependent bending), and 40 post-implant pelvic CT volumes (38 paired with benchmark MRI patients).

2 MATERIALS AND METHODS

2.1 *Patient Cohorts and Public Benchmark*

Table 1 summarizes the data. The primary cohort is 54 intraoperative 3T T2W turbo spin-echo MRI scans acquired during MRI-guided interstitial gynecologic brachytherapy in the AMIGO suite at Brigham and Women's Hospital (Harvard Medical School, Boston, MA, USA) with Institutional Review Board approval [3,4,32]. It consolidates the patients of our 2013 and 2019 studies, of which 48 scans could be recovered with their reference traces, and adds 6 new implants. Expert medical physicists traced 1,063 reference trajectories (9 to 42 per patient): 759 full-depth interstitial catheters from the primary annotation set (`CY`), which we call *long shafts*, and 304 short obturator and partial-field-of-view tracks from the secondary set (`HT`). In-plane pixel size ranges from 0.31 to 0.88 mm and slice thickness from 1.2 to 6.0 mm.

Xiao *et al.* [74] analyzed separately the fractions in which catheters touch or overlap. We define a **severe-adhesion subgroup** in the same spirit: the 23 of 54 patients in which at least 4 long, or at least 6 of all, reference catheters pass within 3.0 mm (2D in-plane Euclidean centerline distance on shared axial slices) of another catheter. All 571 catheters of these patients are evaluated; 191 of them (166 long) lie within 3.0 mm of a neighbor, and 233 do across the whole cohort.

Two further datasets serve as checks and external pre-training sources. All 40 doubly traced scans were contoured independently by two expert teams (`CY+HT`, 723 trajectories = 505 `CY` long + 218 `HT` short/obturator; `LB/AMe`, 513 trajectories, of which 460 are long); the 424 strict long-to-long matched pairs and 476 all-catheter matched pairs within 3.5 mm define the human inter-observer floor, and the de-

TABLE 1 **Clinical imaging cohorts and evaluation protocols.** Primary evaluation uses strictly nested 5 × 5 patient-level out-of-fold (OOF) cross-validation with 100 % reference-free axial cropping (`k_lim_auto`) on 54 clinical 3T T2W MRI datasets from Brigham and Women's Hospital (BWH AMIGO).

Cohort / Subset	Source	Pts	Cath.	XY (mm)	Δz (mm)	Evaluation role
BWH GYN full cohort[a]	BWH (3T T2W)	54	1063	0.31–0.88	1.2–6.0	Nested 5 × 5 OOF benchmark
Long shafts (primary)	BWH (3T T2W)	54	759	0.31–0.88	1.2–6.0	Full-depth active dwell shafts
Short / obturator (secondary)	BWH (3T T2W)	54	304	0.31–0.88	1.2–6.0	Partial-FOV & template tracks
Adhesion subgroup[b]	BWH (3T T2W)	23	571	0.31–0.88	1.2–6.0	Touching / crossing shaft test
Dual-observer subset[c]	BWH (3T T2W)	40	723	0.31–0.88	1.2–6.0	Inter-observer floor (`CY+HT`, `LB/AMe`)
External SSL & specificity[d]	MSD + BWH	50 (51)	0	0.47–0.88	1.6–4.8	3D SSL (51) & 0-needle test (48)

[a] Consolidated superset of the cohorts of Pernelle *et al.* (2013, 10 patients) and Zaffino *et al.* (2019, 50 patients), extended with new implants; 9–42 reference trajectories per patient (20 standard-resolution 320 × 256 scans and 34 high-resolution 640 × 416 / 768 × 768 scans).

[b] Severe catheter-adhesion subgroup: the 23/54 patients in which ≥ 4 long or ≥ 6 total reference catheters pass within ≤ 3.0 mm of another catheter on shared axial slices. All 571 catheters of these patients are evaluated; 191 of them (166 long) lie within 3.0 mm of a neighbour (233 across the whole cohort).

[c] All 40/40 dual-traced clinical scenes independently contoured by two expert medical physics teams (`CY+HT`: 723 total trajectories = 505 `CY` long + 218 `HT` short/obturator; `LB/AMe`: 513 trajectories, with 465 consensus-fused with `CY`, 12 matched to `HT` within ≤ 5.0 mm [11 within ≤ 3.5 mm], and 36 unique `LB/AMe` trajectories forming the 759-trajectory union).

[d] Fifty-one external pelvic 3T T2W MRI volumes from 50 distinct subjects: 48 public CC-BY-SA 4.0 volumes from MSD `Task05_Prostate` (0.60–0.75 mm in-plane, 3.0–4.0 mm slices; zero implanted catheters, evaluated for false-positive specificity) plus 3 non-benchmark BWH pelvic T2W volumes from 2 distinct subjects (`Case 061` [2 series, 0.88 mm in-plane, 4.8 mm slices] and `Case 100 Prostate` [1 series, 0.47 mm in-plane, 1.6 mm slices], zero patient overlap with the 54 benchmark cases) used for self-supervised 3D U-Net pre-training.

duplicated union of both teams contains 759 trajectories (723 `CY+HT` + 36 unique `LB/AMe` trajectories). For self-supervised 3D representation learning and zero-needle specificity testing, we use 51 external pelvic 3T T2W MRI volumes from 50 distinct subjects: all 48 pelvic T2W scans of the public Medical Segmentation Decathlon prostate task [81] (`MSD Task05_Prostate`; 0.60 to 0.75 mm in-plane, 3 to 4 mm slices) plus 3 non-benchmark BWH pelvic T2W series from 2 distinct subjects (`Case 061` [2 series, 0.88 mm in-plane, 4.8 mm slices] and `Case 100 Prostate` [1 series, 0.47 mm in-plane, 1.6 mm slices], zero patient overlap with the 54 benchmark cases).

2.2 *Stage 0: Analytical Dark-Core Symmetry Prior and Reference-Free Cropping*

The physical idea is simple. On an axial slice a catheter cross-section is a dark core that is surrounded by brighter tissue in *every* direction, whereas a fascial plane or a muscle border is dark on one side only. Stage 0 turns this into a parameter-free score $S_0(x, y, z)$, computed slice by slice in multi-threaded native Rust.

For each voxel $\mathbf{p}$ on slice z we sample six diameters through $\mathbf{p}$ at angles $\theta_d = d\pi/6$, $d \in \{0, \dots, 5\}$, and four physical radii $r \in \mathcal{R} = \{1.35, 1.85, 2.5, 3.3\}$ mm, converting each radius to pixels with the in-plane spacing. For diameter d and radius r the darker of the two ring samples is kept,

$$m_{d,r} = \min\big(I(\mathbf{p} + r\mathbf{u}_d),\ I(\mathbf{p} - r\mathbf{u}_d)\big),$$

so that a one-sided edge, where one end of the diameter stays dark, produces a small $m_{d,r}$. The six values are sorted, $m_{(1),r} \le \ldots \le m_{(6),r}$, and the ring brightness is a robust weighted order statistic that still tolerates one dark neighbor,

$$S_{\text{ring}}(\mathbf{p}) = \max_{r \in \mathcal{R}} \left[0.45\, m_{(2),r} + 0.55\, m_{(3),r} \right].$$

The response S_0 grows with the positive contrast between this ring brightness and the 3 × 3 smoothed center intensity and with the relative darkness of the core, and is normalized per volume. Candidate centers are the local maxima of S_0 after non-maximum suppression.

Before running Stage 1, a **100% reference-free, tilt-safe axial template locator** (`k_lim_auto`) identifies the caudal entry slice directly from the 3D intensity volume and $S_0(x, y, z)$: it detects the caudal-to-cranial transition from the air/template saturation drop (`p90` intensity transition) and the emergence of clustered dark-core candidates within the central field of view, setting the lower axial crop bound with a mean signed difference of −1.91 slices (median −2.0, IQR 1.0 slice; mean 58.9 active axial slices vs. 57.2 under `.acsv`) relative to clinical scene annotations while never consulting reference trajectories. Placing the crop boundary ~2 slices deeper caudally prevents tilted perineal templates from truncating in-FOV catheter bases (0/1,063 truncated in-FOV bases under `k_lim_auto` vs. 42/1,063 under `.acsv`), preserving 100.0% of in-FOV catheter depth and 99.65% of total unclipped VTK z-span. Stage 0 takes 0.27 s per full volume on average on a 24-core workstation CPU (0.07 s on standard 320×256 scans and 0.38 s on high-resolution 640×416 / 768×768 scans).

2.3 *Stage 1: Self-Supervised and Multi-Rater Multi-Head 3D U-Net*

Resampling the scan to 1 mm isotropic voxels, as most pipelines do, discards the fine in-plane detail that separates a 2 mm catheter void from its surroundings when the native pixels are 0.3 to 0.9 mm. Working on the native grid instead, a standard 3×3×3 convolution mixes information over up to ten times more distance along z than in-plane. Stage 1 (`NeedleUNet3D`, Figure 2a) therefore operates on the native grid, uses 3 × 3 × 3 residual blocks with squeeze-and-excitation attention at four scales (24, 48, 96, and 144 channels), pools only in-plane at the first two scales so that the thick slice axis is not downsampled early, and adds a dilated convolution at the deepest scale. Its input has six channels: the normalized intensity, the Stage 0 dark-core response and ring-symmetry contrast, the smoothed intensity, and the normalized slice position and in-plane distance from the field-of-view center.

The network predicts four outputs:

1. **Tubular mask** P_{mask}: focal binary cross-entropy, weighted 19-fold on catheter voxels and up-weighted on bright Stage 0 responses that are not catheters, plus soft Dice on the tubular label around each reference trajectory.
2. **Centerline peak** P_{peak}: a sharp 2D Gaussian ($\sigma = 0.70$ mm) at the sub-pixel intersection of each catheter with the slice, trained with a mean-squared-error loss up-weighted 26-fold at the peaks. Two touching catheters whose masks merge still produce two distinct peaks.

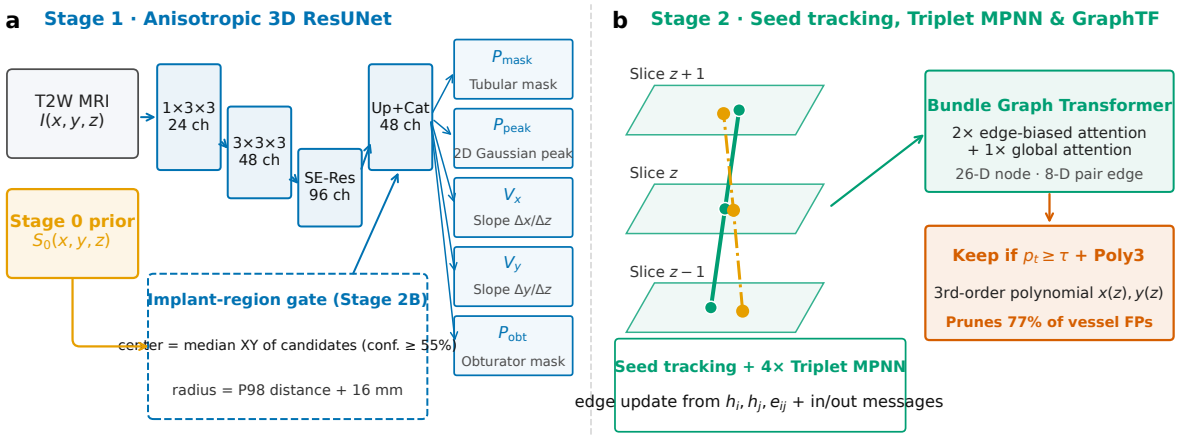


FIGURE 2 **Stage 1 and Stage 2 in detail. (a)** Stage 1 takes the normalized T2W volume and the Stage 0 maps into an anisotropic 3D residual U-Net (`NeedleUNet3D`) pre-trained on 51 external pelvic MRI volumes with synthetic susceptibility tubes and anatomical hard negatives, then fine-tuned with dual-observer consensus supervision to predict four outputs: tubular mask, centerline peak, two direction components, and obturator. **(b)** Stage 2 traces tracks from the seeds in native code, scores their nodes and edges with a 4-layer triplet message-passing network, stitches collinear split fragments across signal dropouts, scores every trajectory in the context of the whole bundle with a graph transformer, localizes each distal tip with a 91-step dilated 1D ResNet + self-attention axial network (`Dedicated1DTipLocalizer`), and smooths each trajectory with a cubic polynomial.

3. **Direction field** $(V_x, V_y) = (\Delta x/\Delta z, \Delta y/\Delta z)$: the local slope of the trajectory, trained with a smooth L_1 loss inside the mask, with an additional term that penalizes slice-to-slice inconsistency of the predicted direction.
4. **Obturator channel** P_{obt}: a dedicated segmentation of the large central obturator and tandem, so that its lumen and rim do not spawn false needles.

To improve feature discrimination on dark pelvic anatomy without leaking test-patient information, `NeedleUNet3D` undergoes two training phases:

– **External pelvic MRI pre-training with synthetic susceptibility tubes and anatomical hard negatives.** The full 3D U-Net (encoder, decoder, and task heads) is pre-trained for 45 epochs across the 51 external pelvic T2W MRI volumes from 50 distinct subjects (48 public `MSD Task05_Prostate` volumes + 3 non-benchmark BWH pelvic T2W volumes, zero patient overlap with the 54 benchmark cases). During each iteration, physics-guided 3D cubic-spline susceptibility tubes (dark core with realistic T2W partial-volume rim, dropout pockets, and variable curvature) are injected into the needle-free pelvic volumes alongside 3D masked volume patches. The network minimizes a joint loss $\mathcal{L}_{SSL} = \mathcal{L}_{tube} + 0.30\,\mathcal{L}_{MAE}$, where $\mathcal{L}_{tube}$ supervises tubular mask, peak, and 3D direction regression on the synthetic tubes while penalizing non-implanted high-symmetry pelvic structures ($S_o > 0.25$: vessels, fascia, obturator internus bor-

ders) as explicit anatomical hard negatives, and $\mathscr{L}_{\text{MAE}}$ reconstructs masked 3D tissue patches (reaching 0.867 validation Dice on synthetic tubes).

– **Multi-rater consensus supervision.** On the 40 dual-annotated training scans (`CY+HT` and `LB/AMe`) within each outer training fold ($n = 43$–44 patients), consensus-matched trajectories ($n = 465$ pairs within ≤ 3.5 mm) are supervised along their inverse-variance precision-weighted centerline (weight 1.15), primary-only (`CY`) trajectories at weight 1.00, secondary-only (`LB/AMe`) trajectories at soft weight 0.55, and secondary cranial tip slices with a soft axial Gaussian tip kernel.

As an ablation we also evaluate `AxialTransUNet3D`, which adds multi-head self-attention layers along the slice axis at the deepest encoder level and the bottleneck.

2.4 *Stage 2: From Slice Detections to a Coherent 3D Bundle*

Stage 2 answers three questions: which detections on neighboring slices belong to the same catheter, which of the resulting 3D curves are real, and exactly where each catheter terminates along the slice axis.

2.4.1 *Stage 2A: Native Seed Tracking, Triplet Message Passing, and Collinear Stitching.*

On each slice, local maxima of P_{peak} and of S_0 become nodes with a 16-dimensional feature vector (normalized position, P_{mask}, P_{peak}, S_0, local 3×3 intensity statistics, predicted slope, and obturator proximity). Directed edges connect nodes across 1 to $\max(4, 24\ \text{mm}/\Delta z)$ slices. An edge is kept when its chord slope is at most 0.56 and its in-plane displacement lies within $2.8 + 0.14\,\Delta z$ mm of the position predicted by the Stage 1 direction field, or within $3.5 + 0.22\,\Delta z$ mm outright. Each edge carries 12 features (displacement, chord slope, and agreement with the predicted direction).

Two consecutive links can each look plausible on their own yet form an abrupt bend when considered together; the network therefore scores each link using information from the links entering and leaving its two nodes. `NeedleSliceMPNN` is a 4-layer triplet-collinearity message-passing network (hidden size 64). Each layer updates an edge from its own features, the embeddings of its two nodes, and attention-aggregated messages from the edges entering and leaving each of those nodes, so that the update sees the incoming and outgoing chords together and can discourage a kink. The network outputs refined sub-pixel offsets $(\delta x_i, \delta y_i)$ and node and edge probabilities.

Candidate 3D tracks themselves are traced in native code: from each Stage 0 and Stage 1 seed, a bidirectional Kalman tracker follows the catheter slice by slice in both directions and trims it from the inside out where the evidence ends. Tracks that run within 1.65 to 2.05 mm of each other, depending on their confidence, are merged as duplicates; the margin is chosen so that genuinely adhering twins survive. Next, **collinear axial split-fragment stitching and extrapolation-aware deduplication** reconnects trajectories broken across 1 to 3 slices by local signal dropouts or vessel crossings: pairs of non-overlapping or slightly overlapping track fragments (axial gap $\leq$ 20 mm, overlap $\leq$ 6 mm) whose linearly extrapolated positions at their shared axial midpoint agree within $\leq$ 2.2 mm with a 3D slope-vector difference $\leq$ 0.14

are merged into a single full-length trajectory (102 stitched trajectories across the 54 cases), while split fragments that lie within $\leq$ 2.1 mm along $\geq$ 72% of a longer high-confidence shaft are absorbed. The surviving tracks are passed to Stage 2B.

2.4.2 *Stage 2B: Bundle Graph Transformer, Dedicated 1D Axial Tip Localizer, and Synthetic Pre-Training.* After Stage 1B, recall is 94.2% (97.0% on long shafts), but each patient still carries 11.6 false tracks on average (626 across 54 patients) along vessels and obturator edges; Stage 2A reduces false tracks by 43% to 6.6 per patient (359 total, lifting precision from 61.5% to 73.5% and F1 from 74.4% to 82.5%). Real catheters have a property these false tracks lack: they form a *bundle*. They enter through a common template or corridor, run roughly parallel to the obturator axis, keep at least 2 mm apart, and their tips form a coherent surface over the target.

Each candidate trajectory is summarized by a list of 26 measurements, its *token* (length, linear-fit residual, 75th-percentile S_o and P_{mask}, P_{peak}, P_{obt}, the three message-passing probabilities, agreement between models, the deviation of its entry position and slope from the bundle median, and the contrast gradient at its cranial end), together with 8 pairwise features between trajectories (minimum 3D clearance, mean in-plane separation, angular divergence, z-overlap, and duplicate indicators). The `TipAwareBundleGraphTransformer` applies two edge-biased attention layers over neighboring trajectories and one global self-attention layer over the whole bundle. It outputs three logits (a recall-oriented score, a strict one-to-one score, and a long-shaft one-to-one score), combined into a confidence p_m, alongside an initial bundle-context tip offset. An implant-region gate inspired by the IRGM of Xiao *et al.* [74] is applied geometrically: the implant center is the median in-plane position of candidates with confidence $\geq$ 55%, the implant radius is the 98th percentile of their distances plus 16 mm, and candidates outside this radius require 22 points higher confidence.

Because a single scalar offset from bundle summary statistics cannot resolve fine sub-slice signal drop-offs or air-pocket artifacts at the cranial end of a catheter, Stage 2B incorporates a **Dedicated 1D Axial Tip Localizer** (`Dedicated1DTipLocalizer`, Figure 2b). Along the cranial-most trajectory tangent of each candidate, the localizer samples a **91-step axial profile** over [−25.0, +20.0] mm at 0.5 mm sub-slice intervals. At every step it extracts **22 multi-scale tubular, intensity, and anatomical features**: center and 3 × 3 / 5 × 5 ring T2W intensities, local annular contrast ($r = 3$ vs. 1, $r = 5$ vs. 1), first- and second-order axial gradients (∂_z, ∂_{zz}), Stage 0 S_o center/max/drop, Stage 1 P_{mask} and P_{peak} center/max/drop, obturator probability P_{obt}, normalized axial coordinate, step indicators, and bundle-relative z-depth. A **3-block dilated 1D ResNet** (hidden width 48, kernel size 5, dilations 1, 2, 4) followed by **4-head 1D Self-Attention** processes the 91-step sequence and drives a dual prediction head: a **91-bin soft-argmax termination probability distribution** combined with a **Smooth-L_1 continuous residual offset head**, supervised with Gaussian target distributions ($\sigma = 1.25$ mm), focal peak classification, and huberized L_1 displacement loss. Each accepted trajectory ($p_m \geq \tau$, default $\tau = 50\%$) has its cranial tip updated by the 1D localizer and is smoothed with a cubic polynomial fit in z (Poly3).

To study zero-shot and low-label bundle scoring, we also built **SynthBrachy3T**, a generator of 300 synthetic candidate-trajectory bundles with the same 26 node and 8 edge features as real candidates (12 to 23 true long shafts per bundle, 30% with an adhering twin 1.65 to 2.75 mm away, 2 to 8 short obturator tracks, 3 to 11 split duplicates, and 4 to 14 vessel-like false tracks). Stage 2B can be pre-trained on SynthBrachy3T and then fine-tuned purely on the clinical training bundles of the current outer fold.

3 EXPERIMENTAL SETUP

3.1 *Strictly Nested Cross-Validation Protocol and Baselines*

Every learned component, every ablation, and every baseline is trained and evaluated under a **strictly nested 5 × 5 patient-level out-of-fold (OOF) protocol** across the 54 patients, eliminating three subtle sources of optimistic bias present in standard stacked evaluations:

1. **Disjoint inner validation for early stopping and calibration.** For each outer fold $f \in \{0, \dots, 4\}$ (43–44 outer-train patients, 10–11 outer-test patients), outer-train patients are partitioned deterministically into an 80% inner-train subset (34–35 patients) and a 20% inner-validation subset (9 patients). Stage 1 U-Net checkpoint selection, Stage 2A MPNN and Stage 2B Graph Transformer early stopping, `Dedicated1DTipLocalizer` model selection, and confidence temperature scaling are performed exclusively on the 9 inner-validation patients; the outer-test fold is touched exactly once at final evaluation.
2. **100% reference-free axial cropping (`k_lim_auto`).** Every outer-test volume is cropped using the automatic `k_lim_auto` template-plane locator derived solely from the raw 3D MRI and Stage 0 S_0 volume, with zero fallback to clinical scene files or ground-truth trajectory coordinates.
3. **Inner-OOF cross-fitting of Stage 1 features (4c).** In a standard 2-stage pipeline (4a, 4b), Stage 2 is trained on outer-train patients using Stage 1 probability maps produced by the outer-fold Stage 1 model (which trained on those same outer-train patients' images). To verify that this causes zero stacking optimism, we also trained 5 × 5 = 25 inner-fold Stage 1 `NeedleUNet3D` models so that every outer-train patient receives strictly out-of-fold Stage 1 probability maps when training Stage 2A, Stage 2B, and `Dedicated1DTipLocalizer` (configuration 4c in Table 2).

We re-implemented three published approaches on the identical outer folds and `k_lim_auto` bounds:

– **PMB 2019: 3D U-Net + DBSCAN** [4]. Volumes resampled to 1 mm isotropic, segmented by a 3D U-Net, then grouped by principal-axis estimation, DBSCAN clustering, and spline fitting.
– **ResUNet + clDice + skeleton** [40,64]. A native-resolution 3D residual U-Net trained with binary cross-entropy, soft Dice, and the centerline-Dice loss, fol-

lowed by morphological skeletonization, connected components, and polynomial centerline fitting.
– **Xiao 2026 Sync-Track** [74]. Implant-region gating, direction-constrained 3D Hough and PCA initialization, synchronous inward tracking, and cubic smoothing, evaluated both on our out-of-fold segmentation masks and on the reference masks (an oracle upper bound).

3.2 *Metrics and Statistics*

Predicted and reference trajectories are matched one-to-one with the Hungarian algorithm (SciPy `linear_sum_assignment`): pairs whose mean in-plane distance over the shared slices exceeds 3.5 mm are forbidden before solving, so the assignment first maximizes the number of valid pairs and then minimizes their distance. Error distributions are computed over matched pairs; a patient with no matched pair contributes fixed penalty values (3.5 mm shaft, 10 mm curve, 20 mm tip and HD95) to the per-patient arrays used in the statistical tests. We report:

1. **Detection**: recall, precision, and F1 under strict one-to-one matching (plus obturator-adjusted precision P_{adj} and $F1_{\text{adj}}$ when evaluating the 759 long active-dwell shafts alone, so predictions that match a valid secondary short/obturator reference track are not counted as false positives).
2. **Shaft error** E_{shaft}: the in-plane distance between the two curves at 40 evenly spaced positions along their shared length, summarized by mean, standard deviation, median, interquartile range, and 95th percentile.
3. **3D curve distance** D_{avg} and **95th-percentile Hausdorff distance** HD_{95}, computed symmetrically on 50 resampled points per curve.
4. **Tip error** E_{tip}: the 3D Euclidean distance between the predicted and reference deepest (cranial) tip.

Calibration and reason-coded decisions. Every candidate track in the stitched Stage 2B pool (1,960 candidates across the 54 patients, 36.3 per patient: 1,356 true within $\leq$ 3.5 mm and 604 false) is also treated as a binary prediction: it is a real catheter if its mean in-plane distance to some reference trajectory over their shared slices is at most 3.5 mm, and its predicted probability is the Stage 2B confidence p_m. Over all candidates we report a reliability diagram, the expected calibration error (ECE, ten equal-width bins), and the Brier score; temperature scaling is fitted on the inner-validation split of each fold and applied to the outer-test fold. Each decision of the decoder carries a typed reason code (kept, below threshold, duplicate of a kept track, outside the implant region, low dark-core symmetry), and for every discarded candidate that is a real catheter we record whether that catheter is already covered by a kept track (redundant) or lost.

Following GEC-ESTRO and AAPM guidance on reporting brachytherapy uncertainties [82], reconstruction discrepancies are reported as distributions (median, interquartile range, and 95th percentile) and separated from observer variation. Differences between methods are tested over the 54 patients with paired two-sided Wilcoxon signed-rank tests, and 95% confidence intervals come from 1,000 patient-level bootstrap resamples.

3.3 *Computing Environment, Software, and Code*

Training and the nested 5 × 5 benchmarks ran on a Linux workstation (Arch Linux) with a 24-core AMD Ryzen Threadripper PRO 7965WX (48 logical threads), 256 GB of RAM, and two NVIDIA GPUs (GeForce RTX 4090, 24 GB, and RTX 4070 Ti SUPER, 16 GB) using Python 3.12/3.13, PyTorch 2.4/2.11, SciPy 1.16, NumPy 2.4, and ONNX Runtime 1.23. The Stage 0 kernel is written in Rust (crate `needlefinder_rust`, multi-threaded with the standard library, exposed to Python through a C interface and `ctypes`, compiled in release mode with link-time optimization). For clinical deployment, the trained Stage 1, Stage 2B, and 1D tip localizer networks are exported to ONNX and run with ONNX Runtime inside an extension for 3D Slicer 5.10 [83]; no GPU is required.

The 3D Slicer extension, the Rust kernel, and the exported models are available at https://github.com/needlefinder/NeedleFinder. All training, nested cross-validation, statistical analysis, SynthBrachy3T, and figure/table generation scripts are archived in `vault/paper/benchmarks`.

4 RESULTS

4.1 *Accuracy Against Published Methods and Ablation Analysis*

Table 2 reports the complete benchmark. Part A compares all methods on the whole cohort (54 patients, 1,063 catheters), on the 759 full-depth active-dwell interstitial shafts (`CY`), and on the severe-adhesion subgroup (23 patients, 571 catheters); Part B reports the step-by-step architectural ablations and threshold sweep on the same nested folds. Figure 3 shows the full per-catheter error distributions.

Five findings stand out.

Voxel segmentation alone over-fragments. The ResUNet + clDice baseline finds almost every catheter (92.4% recall) and is accurate where it matches (0.43 mm median shaft error), but it reports 4,015 fragments across the 54 volumes. Its precision is 24.5% and its F1 38.7%. Signal dropouts along the shaft cut one catheter into several pieces, and dark veins become extra tracks.

Structure-aware tracking needs a clean mask. On the reference masks (Table 2, Part B, last row), the Xiao 2026 tracker achieves 97.7% recall, 90.0% F1, and 0.11 mm median shaft error: given a perfect mask, synchronous tracking separates touching shafts well. On out-of-fold MRI masks its F1 drops to 54.2% (precision 39.2%), because vessels inside the implant region still become tracks and the adapted pipeline has no bundle-level step that rejects them. Tracking alone does improve on skeletonization, by 15.5 F1 points on the whole cohort.

NeedleFinder 2026 achieves the highest F1 and lowest tip error across all evaluated methods. At the default threshold $\tau = 50\%$, `NF 2026 + Synth Brachy3T + 1DTip (4b)` reaches **91.4% recall** (972/1,063), **86.9% precision** (972/1,118), and **89.1% F1** (rising to **89.4% F1** at $\tau = 55\%$, **89.2% F1** at $\tau = 60\%$, and **93.6% recall** at $\tau = 35\%$ or **94.5% recall** at $\tau = 25\%$). The gains over all three published baselines are highly statistically significant: **+6.0 F1 points** over PMB 2019 (89.1% vs. 83.1%, $p = 1.1 \times 10^{-5}$), **+34.9 F1 points** over Xiao 2026 ($p =$

TABLE 2 **Catheter reconstruction accuracy on 54 BWH patients (1 063 catheters), strictly nested 5 × 5 patient-level out-of-fold.** Part A compares end-to-end methods on the whole cohort, the 759 long interstitial shafts, and the severe catheter-adhesion subgroup; Part B ablates **NeedleFinder 2026** stages, methodological upgrades (Items 1–5), and ground-truth-mask tracking on identical folds. Bold: best; underline: second best within each block. Shaded: **NeedleFinder 2026** variants (pre-specified $\tau = 50\,\%$).

	Detection, 1-to-1 (%)			E_{shaft} (mm)			E_{tip} (mm)		Time (s)
Method	Rec.	Prec.	F1	Mean	Median	P95	Mean	Median	CPU
Part A — End-to-end comparison									
Whole cohort (54 patients, 1 063 catheters)									
PMB 2019 3D U-Net[a]	88.8	78.0	83.1	**0.52**	**0.42**	**1.11**	10.75	3.39	14.80
ResUNet + clDice + Skel.[b]	92.4	24.5	38.7	0.55	0.43	1.36	10.08	3.10	3.85
Xiao 2026 Sync-Track[c]	87.7	39.2	54.2	0.66	0.47	1.82	15.65	6.04	**3.42**
NeedleFinder 2026 (no SynthBrachy3T)[d]	**92.5**	85.3	88.7	0.59	0.47	1.35	6.39	2.26	4.15
NeedleFinder 2026 + Synth + 1DTip[e]	91.4	**86.9**	**89.1**	0.59	0.47	1.37	**6.23**	2.27	4.15
NeedleFinder 2026 + Inner-OOF Cross-Fit[e]	92.3	86.1	**89.1**	0.59	0.47	1.37	6.25	**2.24**	4.15
Long interstitial shafts (54 patients, 759 catheters, obturator-adjusted precision)[f]									
PMB 2019 3D U-Net	86.7	87.6	87.2	**0.52**	**0.40**	**1.23**	14.93	4.04	–
ResUNet + clDice + Skel.	91.3	51.5	65.8	0.57	0.43	1.40	14.67	3.53	–
Xiao 2026 Sync-Track	84.2	53.7	65.6	0.64	0.46	1.84	16.02	6.90	–
NeedleFinder 2026 (no SynthBrachy3T)	**93.4**	87.1	90.2	0.56	0.45	1.24	7.47	2.49	–
NeedleFinder 2026 + Synth + 1DTip	91.3	**88.7**	90.0	0.57	0.45	1.27	7.41	**2.41**	–
NeedleFinder 2026 + Inner-OOF Cross-Fit	92.5	88.6	**90.5**	0.56	0.45	1.27	**7.17**	2.44	–
Severe catheter-adhesion subgroup (23 patients, 571 catheters)[g]									
PMB 2019 3D U-Net	94.6	79.3	86.3	0.54	0.43	1.22	11.88	3.70	–
ResUNet + clDice + Skel.	**96.0**	32.3	48.4	**0.52**	**0.42**	**1.18**	11.58	3.37	–
Xiao 2026 Sync-Track	91.6	49.0	63.9	0.64	0.47	1.83	17.69	7.52	–
NeedleFinder 2026 (no SynthBrachy3T)	94.6	87.2	90.8	0.58	0.45	1.36	**7.03**	2.48	–
NeedleFinder 2026 + Synth + 1DTip	93.2	**89.6**	91.3	0.58	0.46	1.41	7.08	2.51	–
NeedleFinder 2026 + Inner-OOF Cross-Fit	94.6	89.1	**91.8**	0.57	0.46	1.34	7.30	**2.47**	–
Part B — Controlled ablation & methodological progression (whole cohort, 1 063 catheters)									
Stage 0: Dark-core prior (`k_lim_auto`)	82.8	69.1	75.3	0.60	0.47	1.41	9.91	3.08	**0.27**
Stage 1B: SSL + Multi-Rater U-Net + tubes	**94.2**	61.5	74.4	**0.57**	**0.46**	1.31	9.29	2.95	2.84
Stage 2A: Triplet MPNN only	93.9	73.5	82.5	0.58	0.47	1.31	8.95	2.83	4.09
+ Item 1: Nested CV + res. calibration	90.9	83.6	87.1	**0.57**	**0.46**	**1.30**	7.39	2.70	4.12
+ Item 3: Multi-rater Stage 2 supervision	91.7	81.7	86.4	0.58	**0.46**	1.32	7.38	2.57	4.12
+ Item 4a: Collinear split-fragment stitch	91.4	86.9	89.1	0.59	0.47	1.36	6.92	2.56	4.13
+ Item 4b: 1D Tip Localizer ($\tau = 35\,\%$)	93.6	83.4	88.2	0.59	0.47	1.37	6.71	2.32	4.15
+ Item 4b: 1D Tip Localizer ($\tau = 50\,\%$)	91.4	86.9	89.1	0.59	0.47	1.37	**6.23**	2.27	4.15
+ Item 4b: 1D Tip Localizer ($\tau = 55\,\%$)	91.3	**87.6**	**89.4**	0.59	0.47	1.37	6.33	2.27	4.15
Xiao 2026 on GT masks[h]	97.7[r]	83.5[r]	90.0[r]	0.20[r]	0.11[r]	0.83[r]	8.94[r]	1.68[r]	0.12[r]

Rec./Prec./F1: strict one-to-one Hungarian matching ($\leq$ 3.5 mm shaft threshold); E_{shaft}: in-plane centerline distance on matched slices; E_{tip}: 3D distal-tip distance (P95 on whole cohort: 27.27 mm for **NeedleFinder 2026** + Synth + 1DTip vs. 52.70 mm for PMB 2019, 45.76 mm for clDice, 62.69 mm for Xiao 2026).

[a] Adaptation of Zaffino *et al.* (2019): $1 \times 1 \times 1$ mm^3 isotropic 3D U-Net, retrained on our 5-fold splits, followed by slice-peak extraction, collinear chaining, and polynomial fitting.

[b] Native-resolution 3D ResUNet trained with Dice + focal + soft-skeleton clDice loss (2021), followed by distance-transform ridge extraction and inter-slice chaining.

[c] Adaptation of Xiao *et al.* (2026): geometric implant-region gate, connected-component and distance-transform candidates with local PCA direction initialisation, synchronous cost-matrix inward tracking, and 3rd-order polynomial fitting on OOF masks.

[d] Clean nested **NeedleFinder 2026** pipeline (Items 1–5) trained on clinical folds without `SynthBrachy3T` synthetic bundle pre-training.

[e] Full clean nested **NeedleFinder 2026** pipeline with Stage 2B pre-trained on `SynthBrachy3T` and Stage 2C Dedicated 91-step (0.5 mm) 22-channel 1D Axial Tip Localizer (whole-cohort F1: $p = 1.1 \times 10^{-5}$ vs. PMB 2019, $p = 5.2 \times 10^{-10}$ vs. clDice, $p = 3.0 \times 10^{-10}$ vs. Xiao 2026, $p = 0.32$ vs. no-`SynthBrachy3T`, $p = 0.80$ vs. Inner-OOF Cross-Fit; E_{tip}: $p = 1.7 \times 10^{-8}$ vs. PMB 2019, $p = 6.2 \times 10^{-10}$ vs. Xiao 2026).

[f] Evaluated on the 759 full-depth `CY` interstitial shafts; predictions matching the 304 auxiliary `HT` short obturator tracks are

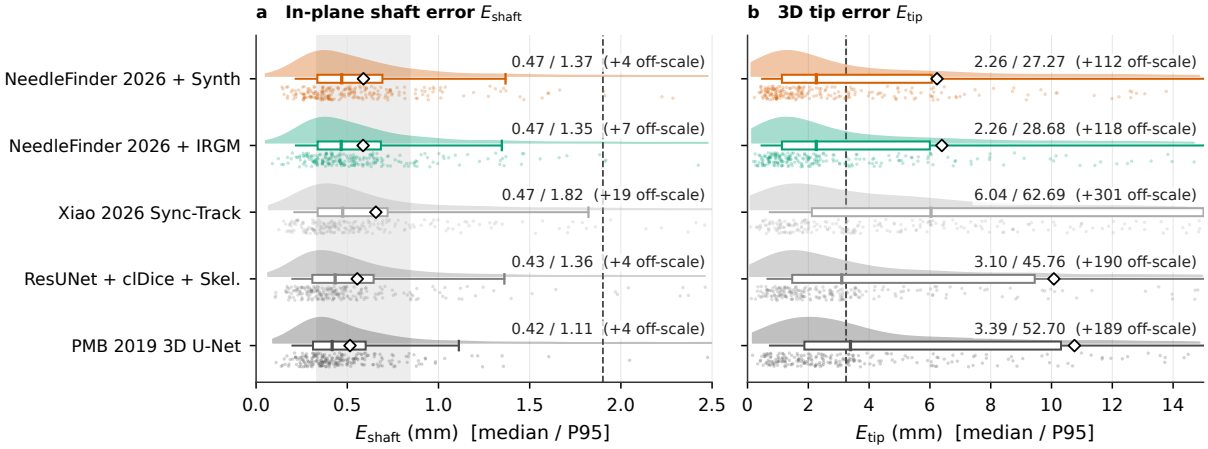


FIGURE 3 **Per-catheter error distributions on the whole cohort** (54 patients; each method's matched catheters). **(a)** In-plane shaft error and **(b)** 3D tip error. Each raincloud shows the density (top), the median and interquartile range (white box), the 5th to 95th percentile (capped bar), the mean (white diamond), and individual catheters (points). Numbers at the right give median / 95th percentile in mm, computed on the full distribution, and the number of catheters beyond the plotted range. In (a) the grey band and dashed line mark the human inter-observer long-shaft interquartile range (0.34 to 0.84 mm) and 95th percentile (1.90 mm); in (b) the dashed line marks the human inter-observer median tip discrepancy (3.24 mm).

3.0×10^{-10}), and **+50.4 F1 points** over ResUNet + clDice ($p = 5.2 \times 10^{-10}$). On matched catheters, the in-plane shaft error is 0.59 ± 0.40 **mm mean** (0.47 **mm median**, 0.36 mm IQR, 1.37 **mm P95**), the 3D symmetric curve distance D_{avg} is 3.20 ± 4.87 mm mean (1.00 mm median), and HD_{95} is 15.36 ± 22.87 mm mean (3.11 mm median). Most notably, the Dedicated 1D Axial Tip Localizer and collinear fragment stitching reduce 3D distal tip error to 6.23 ± 10.49 **mm mean** (2.27 **mm median**, 4.93 mm IQR, 27.27 **mm P95**), significantly outperforming PMB 2019 (10.75 mm mean / 3.39 mm median, $p = 1.7 \times 10^{-8}$), ResUNet + clDice (10.08 mm mean / 3.10 mm median, $p = 2.3 \times 10^{-7}$), and Xiao 2026 (15.65 mm mean / 6.04 mm median, $p = 6.2 \times 10^{-10}$). Nearly three-fifths (**58.7%**, 571/972) of matched catheters have a 3D tip error ≤ 3.0 mm (within roughly one slice thickness), while **16.7%** (162/972) exceed 10 mm. Across the 54 patients, the method misses 91 reference trajectories (1.7 per patient) and reports 146 unmatched tracks (2.7 per patient). On the **759 long interstitial shafts** (Table 2, Part A, middle block), `NF 2026` reaches **91.3% recall** (693/759), **88.7% obturator-adjusted precision** (79.1% strict long-only precision), and **90.0%** $F1_{adj}$ (84.8% strict F1) at $\tau = 50\%$—and **94.7% recall** (719/759) at $\tau = 35\%$—with 0.57 mm mean (0.45 mm median) shaft error and 2.41 mm median (7.41 mm mean) tip error.

Strict 5 × 5 inner out-of-fold cross-fitting confirms zero stacking optimism, and clarifies the exact role of SynthBrachy3T. Training Stage 2 on strictly out-of-fold inner Stage 1 predictions (4c, `NF 2026 + SynthBrachy3T + Inner-OOF Cross-Fit`) yields **92.3% recall** (981/1,063), **86.1% precision**, and **89.1% F1** (0.59 mm mean / 0.47 mm median shaft error; 6.25 mm mean / 2.24 mm median

tip error), and achieves the highest obturator-adjusted F1 on long shafts (**90.5%** $F1_{adj}$, 92.5% recall, 7.17 mm mean tip error) and on the severe-adhesion subgroup (**91.8% F1**, 94.6% recall, 89.1% precision). Furthermore, once the self-supervised multi-rater Stage 1, collinear fragment stitching, and `Dedicated1DTipLocalizer` are in place, our purely clinical pipeline without `SynthBrachy3T` (4a) already reaches **88.7% F1** (92.5% recall, 85.3% precision; 90.2% $F1_{adj}$ on long shafts) and 6.39 **mm mean** / 2.26 **mm median tip error**. At the full 43-patient training budget, adding `SynthBrachy3T` (4b) provides a modest refinement to 89.1% F1 (+0.41 pp, $p = 0.32$) and 6.23 mm mean / 27.27 mm P95 tip error ($p = 0.17$); as shown below, the primary detection gain from `SynthBrachy3T` lies in the zero-clinical-label ($k = 0$) regime.

The advantage is largest when catheters touch. On the 23 severe-adhesion patients (571 catheters), `NF 2026` reaches **91.3% F1** (**93.2% recall**, **89.6% precision**, 0.58 mm mean / 0.46 mm median shaft error, 7.08 mm mean / 2.51 mm median tip error) under 4b and **91.8% F1** (**94.6% recall**, **89.1% precision**, 0.57 mm mean / 0.46 mm median shaft error, 7.30 mm mean / 2.47 mm median tip error) under inner-OOF cross-fitting (4c), compared with 90.8% without synthetic pre-training (4a), 86.3% for PMB 2019 (F1 $p = 0.0012$, tip $p = 4.5 \times 10^{-6}$), 63.9% for Xiao 2026 (F1 $p = 4.8 \times 10^{-7}$, tip $p = 2.4 \times 10^{-7}$), 48.4% for ResUNet + clDice (F1 $p = 4.8 \times 10^{-7}$, tip $p = 7.2 \times 10^{-7}$), and 75.3% for Stage 0 (F1 $p = 4.8 \times 10^{-7}$, tip $p = 2.4 \times 10^{-7}$).

4.2 *Contribution of Each Stage and Leak-Free Learning From Few Patients*

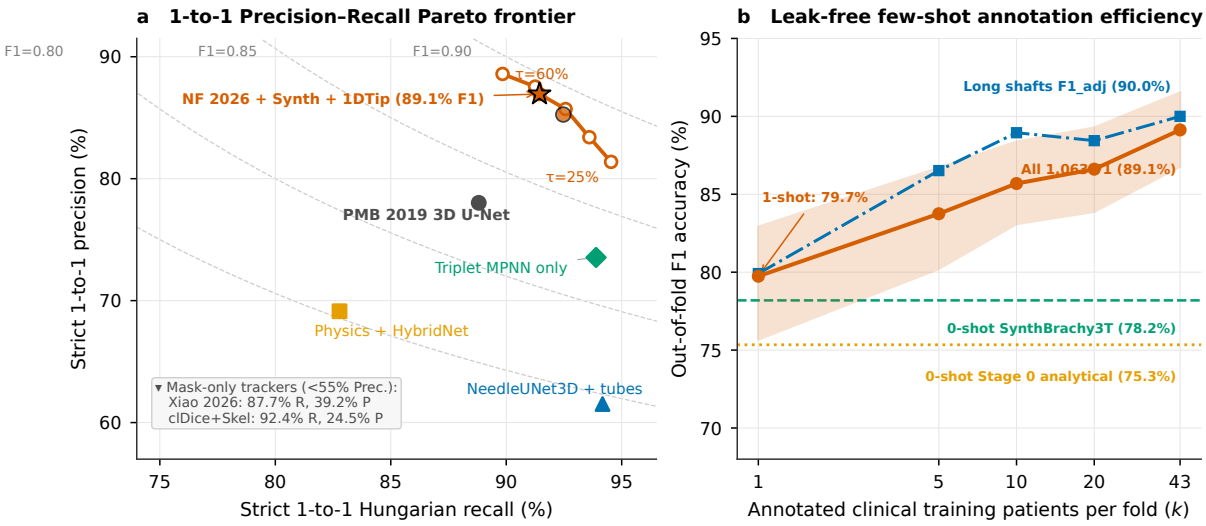


FIGURE 4 **(a) Precision versus recall** for each pipeline stage, each baseline, and the NeedleFinder 2026 threshold sweep on the whole cohort. Dashed curves are lines of constant F1; the star marks the default threshold $\tau = 50\%$ (89.1% F1) and the grey circle next to it is the pipeline without synthetic pre-training (4a, 88.7% F1). **(b) Leak-free Stage 2B clinical support curve** showing accuracy versus number of annotated outer-train patients per fold used to fine-tune the bundle-scoring stage, $k \in \{0, 1, 5, 10, 20, 43\}$, with zero blending from any full-fold clinical model. Shaded bands show the patient-level bootstrap 95% confidence intervals of F1.

Figure 4(a) traces the progression across stages and methodological upgrades on the identical nested folds (Table 2, Part B). Analytical `Stage 0: Dark-core prior (k_lim_auto)` alone achieves **82.8% recall**, **69.1% precision**, and **75.3% F1** (0.60 mm mean / 0.47 mm median shaft error, 9.91 mm mean / 3.08 mm median tip error; 95.3% recall on long shafts). `Stage 1B: SSL + Multi-Rater U-Net + tubes` raises recall to **94.2%** (97.0% on long shafts) with **61.5% precision** (**74.4% F1**) and cuts shaft error to 0.57 mm mean / 0.46 mm median and tip error to 9.29 mm mean / 2.95 mm median. `Stage 2A: Triplet MPNN only` maintains **93.9% recall** while lifting precision by +12.0 pp to **73.5%** (**82.5% F1**, 0.58 mm mean / 0.47 mm median shaft error, 8.95 mm mean / 2.83 mm median tip error). Adding Stage 2B bundle graph-transformer pruning with `+ Item 1: Nested CV + res. calibration` reaches **87.1% F1** (90.9% recall, 83.6% precision, 7.39 mm mean / 2.70 mm median tip error); `+ Item 3: Multi-rater Stage 2 supervision` reaches **86.4% F1** (91.7% recall, 81.7% precision, 7.38 mm mean / 2.57 mm median tip error); `+ Item 4a: Collinear split-fragment stitch` lifts precision to **86.9%** and F1 to **89.1%** (91.4% recall, 6.92 mm mean / 2.56 mm median tip error); and `+ Item 4b: 1D Tip Localizer` at $\tau = 50\%$ reaches **89.1% F1** (88.7% without `SynthBrachy3T`, `4a`; **89.4% F1** at $\tau = 55\%$) while cutting tip error to 6.23 **mm mean** / 2.27 **mm median**.

Figure 4(b) evaluates **strictly leak-free Stage 2B clinical label efficiency**: for each support budget $k \in \{0, 1, 5, 10, 20, 43\}$, the Stage 2B bundle graph transformer is fine-tuned exclusively on `outer_train[:k]` clinical patient bundles of that fold (with zero score blending from any full-fold $k = 43$ clinical model). With zero clinical bundle labels ($k = 0$), analytical Stage 0 alone gets **75.3% F1** (77.1% $F1_{\text{adj}}$ on long shafts), while the **zero-shot SynthBrachy3T-pretrained Stage 2B** (`0_synth_zeroshot`, $k = 0$) reaches **78.2% F1** (+2.85 pp overall, and 83.4% $F1_{\textbf{adj}}$ [+6.3 pp] with 89.9% recall on long shafts, 9.18 mm mean / 3.03 mm median tip error). Fine-tuning Stage 2B on a single annotated clinical patient ($k = 1$) reaches **79.7% F1** (91.4% recall, 79.9% long $F1_{\text{adj}}$, 7.13 mm mean / 3.10 mm median tip error); **5 annotated patients ($k = 5$) reach 83.8% F1** (86.5% **long** $F1_{\textbf{adj}}$**,** 6.89 **mm mean** / 2.49 **mm median tip error), already surpassing the fully supervised 43-patient PMB 2019 3D U-Net baseline (83.1% F1)**; 10 patients ($k = 10$) reach **85.7% F1** (89.0% long $F1_{\text{adj}}$, 7.04 mm mean / 2.60 mm median tip error); 20 patients ($k = 20$) reach **86.6% F1** (88.4% long $F1_{\text{adj}}$, 6.98 mm mean / 2.64 mm median tip error); and all 43 patients ($k = 43$) reach **89.1% F1** (90.0% long $F1_{\text{adj}}$, 6.23 mm mean / 2.27 mm median tip error).

Figure 5 illustrates the stages on two difficult out-of-fold test cases (Case 017 with 23 predicted trajectories vs. 26 reference, and Case 031 with 20 predicted vs. 23 reference). Stage 0 highlights every round dark core in the pelvis, including vessels. Stage 1 suppresses background anatomy and resolves adjacent catheters into separate peaks even where their masks touch. Stage 2 then stitches dropouts, prunes false tracks, localizes the cranial tips, and outputs smooth 3D bundles.

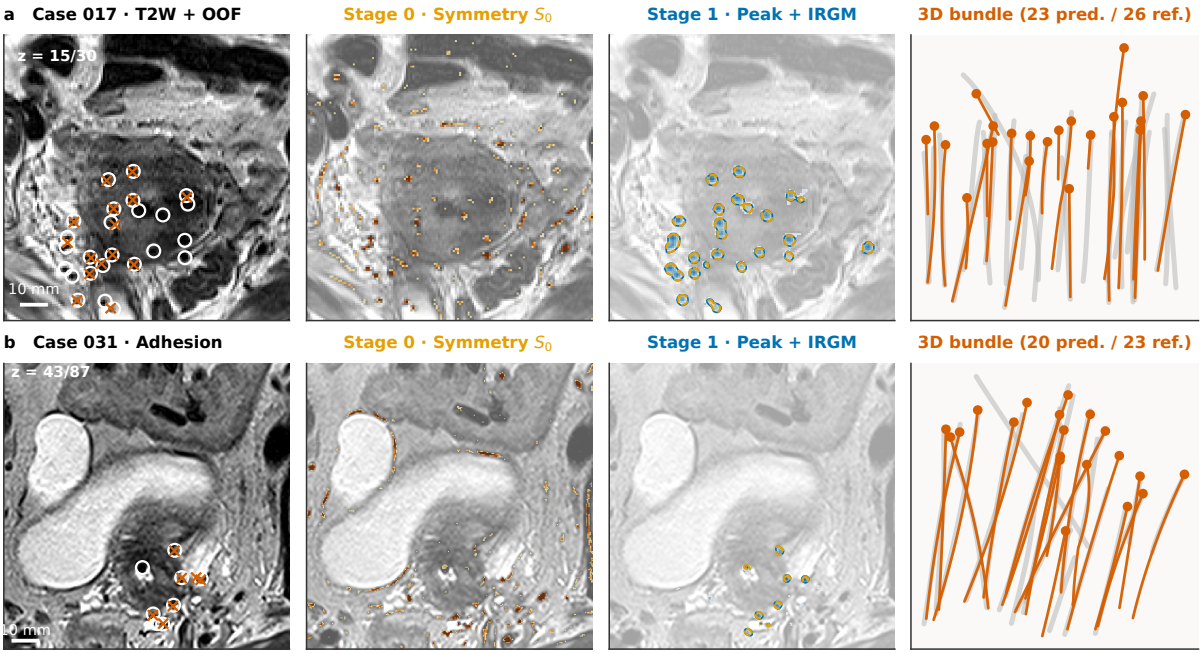


FIGURE 5 **Two nested out-of-fold test cases. (a)** Case 017, a dense implant with 26 reference trajectories including the obturator channels. **(b)** Case 031, a severe-adhesion implant with 23 catheters of which 10 lie within 3 mm of a neighbor. Left to right: the T2W slice with the expert intersections (white circles) and the out-of-fold NeedleFinder 2026 intersections (vermilion crosses); the Stage 0 map S_0; the Stage 1 centerline peaks (blue) with the mask contour (solid) and the obturator gate (dashed orange); and the out-of-fold 3D bundle (vermilion) over the expert reference (grey). Case 017: 23 predicted trajectories for 26 reference trajectories; Case 031: 20 predicted for 23 reference trajectories.

4.3 *Agreement With Experts Compared With Agreement Between Experts*

Because a catheter is only a few pixels wide and slices are thick, two experts clicking centerlines never agree perfectly; inter-observer discrepancies of about 0.6 mm have been reported for breast implants, and more than a third of prostate needles differ by over 1 mm between observers [84,85], and algorithms within that range are considered clinically comparable to a second observer [38,58]. Table 3 measures this across **all 40 doubly traced clinical scenes** (`CY+HT`, 723 trajectories = 505 `CY` long + 218 `HT` short/obturator; `LB/AMe`, 513 trajectories, of which 460 are long) and compares it with `NeedleFinder 2026` evaluated out-of-fold on the exact same 40 scenes.

Comparing the two human teams on long shafts (`LB/AMe` 460 long vs. `CY` 505 long), 424 pairs match one-to-one within ≤ 3.5 mm (**84.0% recall**, **92.2% strict precision** [93.6% obturator-adjusted], **87.9% strict F1** [88.5% obturator-adjusted]), with a shaft discrepancy of 0.70±0.58 **mm mean** (0.49 **mm median**, 0.50 mm IQR, 1.90 **mm P95**) and a 3D tip discrepancy of 6.28±7.29 **mm mean** (3.24 **mm median**, 6.02 mm IQR, 21.88 mm P95). Across all 723 `CY+HT` reference trajectories, `LB/AMe` matches 476 pairs (65.8% recall, 92.8% precision, 77.0% F1; 0.73 ± 0.61 mm mean / 0.50 mm median shaft discrepancy, 6.30 ± 8.02 mm mean / 3.21 mm median tip discrepancy).

TABLE 3 **Human inter-observer agreement versus NeedleFinder 2026 across all 40/40 dual-annotated clinical 3T MRI scenes.** Expert teams CY+HT (n = 723 total trajectories: 505 CY long + 218 HT short/obturator) and LB/AMe (n = 513, of which 460 are long) digitized the same 40 volumes independently; **NeedleFinder 2026** rows are strictly nested 5 × 5 out-of-fold predictions on the same 40 scenes.

	1-to-1 Detection (≤ 3.5 mm)				E_{shaft} (mm)				E_{tip} (mm)	
Comparison pair (eval. vs. reference)	n_{match}	Rec.	Prec.	F1	Mean	SD	Median	P95	Mean	Median
Human inter-observer agreement										
LB/AMe vs. CY long (460 vs. 505)[a]	424	84.0	**92.2**	**87.9**	0.70	0.58	0.49	1.90	6.28	3.24
LB/AMe vs. CY+HT all (513 vs. 723)	476	65.8	92.8	77.0	0.73	0.61	0.50	2.09	6.30	3.21
Automated out-of-fold predictions vs. observer teams across all 40 scenes										
NeedleFinder 2026 (τ = 50 %) vs. CY+HT (768 vs. 723)	653	90.3	85.0	87.6	**0.58**	**0.39**	**0.47**	1.33	**5.26**	**2.10**
NeedleFinder 2026 (τ = 35 %) vs. CY+HT (825 vs. 723)	669	**92.5**	81.1	86.4	0.59	0.40	**0.47**	**1.32**	5.54	2.20
NeedleFinder 2026 (τ = 50 %) vs. union CY+HT+LB (768 vs. 759)[b]	669	88.1	87.1	87.6	0.59	0.40	**0.47**	1.36	5.44	2.17
NeedleFinder 2026 (τ = 50 %) vs. LB/AMe (768 vs. 513)[c]	465	90.6	60.5	72.6	0.77	0.62	0.51	2.11	7.87	3.97

Grubbs' three-cornered-hat intrinsic error SD (n = 432) & Pelvic CT adjudication (n = 70)[d]									
Source (3-way matched)	n 3W Sh.	3W Tip	Role	σ_{RMS}	σ_{MAD}	$\sigma_{z,MAD}$	$\sigma_{z,W}$	CT	Win
Human Rater 1 (CY primary)	432	Ref.	Ref. R1	**0.39**	**0.35**	**0.00**[d]	**3.77**	4.31	–
NeedleFinder 2026 (τ = 50 %, OOF)	432	**0.43**	**2.33**	Eval.	0.56	0.39	2.98	4.94	**4.20**	**78.6 %**
Human Rater 2 (LB/AMe secondary)	432	0.49	3.18	Comp. R2	0.86	0.48	3.55	5.09	8.00	–

One-sided non-inferiority tests vs. human inter-observer discrepancy (N = 40, n = 432)[e]										
Endpoint (mm)	Δ	NF	LB	Diff.	CI lo	CI hi	Pass	p_{boot}	p_{pt}	p_{cath}
In-plane shaft mean (E_{shaft})	+0.25	**0.56**	0.72	−0.16	−0.31	−0.02	Yes	$< 10^{-4}$	1.3e−8	1.6e−41
In-plane shaft median (E_{shaft})	+0.25	**0.43**	0.49	−0.06	−0.19	+0.01	Yes	$< 10^{-4}$	8.2e−10	1.6e−41
3D distal tip median (E_{tip})	+1.00	**2.33**	3.18	−0.85	−1.46	−0.52	Yes	$< 10^{-4}$	2.6e−7	2.2e−14
3D distal tip mean (E_{tip})	+1.50	**5.51**	6.10	−0.60	−1.80	+0.57	Yes	0.0014	1.1e−4	6.8e−19

[a] Strict long-shaft inter-observer comparison (424 matched pairs out of 460 LB/AMe long vs. 505 CY long: 83.96 % recall, 92.17 % strict precision [93.6 % obturator-adjusted], 87.88 % strict F1 [88.5 % obturator-adjusted], 0.70 ± 0.58 mm shaft error [median 0.49 mm, IQR 0.50 mm, P95 1.90 mm] and 6.28 ± 7.29 mm 3D tip error [median 3.24 mm, IQR 6.02 mm, P95 21.88 mm]).

[b] Evaluated against the de-duplicated union of both expert teams (723 CY+HT + 36 unique LB/AMe trajectories = 759 union catheters), showing that 16 apparent false positives against CY+HT alone are genuine catheters traced by LB/AMe (precision rises from 85.0 % to 87.1 %).

[c] Because LB/AMe primarily traced long interstitial shafts (513 traces vs. 723 for CY+HT), unadjusted precision against LB/AMe alone penalizes valid short obturator predictions; recall is 90.6 %.

[d] Reference-free Grubbs' (1948) three-cornered-hat intrinsic source error SD across the n = 432 three-way matched catheters ('3-Way Shaft' and '3-Way Tip': median pairwise error vs. CY in mm; 'Shaft RMS' σ: classic MLE SD; 'Shaft MAD' σ: robust MAD-scaled SD; 'Tip MAD' σ_z: robust MAD axial tip SD, where Rater 1 reaches the 0.001 mm boundary clamp [95 % CI 0.001–2.02 mm] because median signed Δz_{R1-NF} = 0.00 mm; 'Tip Wins.' σ_z: 15 % Winsorized axial tip SD; 'CT Med.' and 'Closer CT': median 3D tip error against independent post-implant Pelvic CT across the n = 70 catheters with $|\Delta z_{CY-LB}| \geq 5.0$ mm on MRI, where **NeedleFinder 2026** is closer to CT than single-rater MRI on 78.6 % [55/70] of disputed tips).

[e] One-sided non-inferiority tests ($H_0 : E_{NF,CY} - E_{LB,CY} \geq \Delta$) at α = 0.05 across the N = 40 dual-traced patients (n = 432 three-way matched catheters), reporting 10,000-resample patient-clustered bootstrap 90 % CIs, patient-level paired Wilcoxon p_{pt} (N = 40), and catheter-level paired Wilcoxon p_{cath} (n = 432; patient-clustered bootstrap p_{boot} = 0.0014 for tip mean, $< 10^{-4}$ for all other endpoints).

Evaluated out-of-fold across all 40 scenes, `NeedleFinder 2026` (τ = 50%, 768 predicted trajectories) matches **653 of 723 `CY+HT` trajectories (90.3% recall, 85.0% precision, 87.6% F1**; rising to **92.5% recall** at τ = 35%) with a mean shaft error of 0.58±0.39 **mm** (0.47 **mm median**, 0.35 mm IQR, 1.33 **mm P95**) and a 3D tip error of 5.26 ± 8.78 **mm mean** (2.10 **mm median**, 4.11 mm IQR, 19.99 mm P95)—within the inter-observer disagreement between the two human teams in mean, median, and 95th-percentile error for both shafts and tips. Even more revealingly, when evaluated against the **de-duplicated union of both expert teams** (`CY+HT` ∪ `LB/AMe`, 759 union trajectories), `NeedleFinder 2026` matches **669 trajectories (88.1% recall, 87.1% precision, 87.6% F1**, 0.59 mm mean / 0.47 mm median shaft error, 5.44 mm mean / 2.17 mm median tip error): **16 trajectories that appeared to be false positives against `CY+HT` alone are in fact genuine catheters independently traced by `LB/AMe`**, raising precision by +2.1 pp. Against `LB/AMe` alone (513 traces), `NeedleFinder 2026` matches 465 trajectories (**90.6% recall**, 0.77 mm mean / 0.51 mm median shaft error, 3.97 mm median tip error; unadjusted precision is 60.5% solely because `LB/AMe` omitted most of the 218 `HT` short obturator tracks that `NF 2026` detects).

To separate each source's intrinsic error from the choice of reference team, the third block of Table 3 applies **Grubbs' (1948) three-cornered-hat estimator** across the n = 432 catheters matched three-way among `CY`, `LB/AMe`, and out-of-fold `NeedleFinder 2026`. The intrinsic robust (MAD-scaled) in-plane shaft standard deviations are 0.35 **mm for Rater 1 (`CY`)**, 0.39 **mm for `NeedleFinder 2026`**, and 0.48 **mm for Rater 2 (`LB/AMe`)** (classic MLE RMS σ: 0.39, 0.56, and 0.86 mm). For axial tip termination, robust MAD σ_z is 0.001 mm (boundary clamp, 95% CI 0.001–2.02 mm) for `CY`, 2.98 **mm for `NeedleFinder 2026`**, and 3.55 **mm for `LB/AMe`**, while 15% Winsorized axial tip σ_z is 3.77 **mm (`CY`)**, 4.94 **mm (`NeedleFinder 2026`)**, and 5.09 **mm (`LB/AMe`)**. As detailed in the fourth block of Table 3, **one-sided non-inferiority tests** ($H_o : E_{\mathrm{NF,CY}} - E_{\mathrm{LB,CY}} \geq \Delta$) at α = 0.05 across all N = 40 dual-traced patients (n = 432 three-way matched catheters) confirm that `NeedleFinder 2026` is statistically non-inferior to human inter-observer discrepancy on all four endpoints: shaft mean (Δ = +0.25 mm: 0.56 vs. 0.72 mm, diff −0.16 mm, 90% cluster bootstrap CI [−0.31, −0.02] mm, patient-level Wilcoxon $p_{\mathrm{pt}} = 1.3 \times 10^{-8}$, catheter-level $p_{\mathrm{cath}} = 1.6 \times 10^{-41}$), shaft median ($\Delta$ = +0.25 mm: 0.43 vs. 0.49 mm, diff −0.06 mm, 90% CI [−0.19, +0.01] mm, $p_{\mathrm{pt}} = 8.2 \times 10^{-10}$, $p_{\mathrm{cath}} = 1.6 \times 10^{-41}$), 3D tip median ($\Delta$ = +1.00 mm: 2.33 vs. 3.18 mm, diff −0.85 mm, 90% CI [−1.46, −0.52] mm, $p_{\mathrm{pt}} = 2.6 \times 10^{-7}$, $p_{\mathrm{cath}} = 2.2 \times 10^{-14}$), and 3D tip mean ($\Delta$ = +1.50 mm: 5.51 vs. 6.10 mm, diff −0.60 mm, 90% CI [−1.80, +0.57] mm, $p_{\mathrm{pt}} = 1.1 \times 10^{-4}$, $p_{\mathrm{boot}} = 0.0014$, $p_{\mathrm{cath}} = 6.8 \times 10^{-19}$). Finally, independent adjudication against paired post-implant **Pelvic CT scans** across the n = 70 catheters where `CY` and `LB/AMe` disagreed by ≥ 5.0 mm on MRI shows that `NeedleFinder 2026` achieves a **median 3D tip error of** 4.20 **mm against CT**, compared with 4.31 **mm for `CY`** and 8.00 **mm for `LB/AMe`**, lying closer to CT than single-rater MRI on **78.6% (55/70)** of disputed catheters.

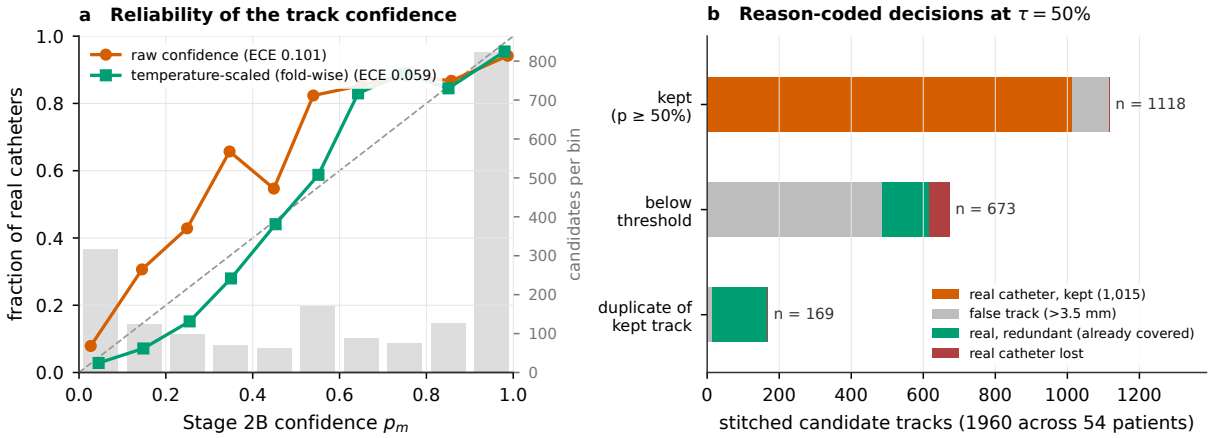


FIGURE 6 **(a) Reliability of the Stage 2B confidence** over all 1,960 stitched candidate tracks of the 54 patients (bars: candidates per bin), raw and after inner-fold temperature scaling. **(b) Reason-coded decisions at the default threshold** $\tau = 50\%$: for each rule, how many candidates are kept or removed and how many of those were false tracks, real catheters already covered by a kept track, or real catheters lost.

4.4 *Confidence Calibration, Reason-Coded Decisions, and Failure Analysis*

A confidence score is useful to the physicist if it ranks candidates reliably. Figure 6(a) shows the reliability diagram over all 1,960 stitched candidate tracks (36.3 per patient: 1,356 true within ≤ 3.5 mm, 604 false): the expected calibration error (ECE) is 0.101 raw (Brier score 0.123) and 0.059 after inner-fold temperature scaling (**Brier score** 0.112; 0.056 **ECE** over the 1,118 kept tracks at $\tau = 50\%$, which have a mean predicted confidence of 91.9%, closest-GT precision of 90.8% [1,015/1,118], and 1-to-1 Hungarian precision of 86.9% [972/1,118]). Figure 6(b) breaks down the reason-coded decisions of the Stage 2B decoder across the 1,960 stitched candidates at $\tau = 50\%$: 1,118 **are kept** (1,015 real [972 1-to-1 matched + 43 near-matched], 103 false > 3.5 mm), 673 **are pruned as `below_threshold`** (487 false [80.6% of all false candidates pruned here], 129 redundant real fragments already covered by a kept track, and 57 real catheters lost), and 169 **are suppressed as `duplicate_of_kept_track`** (153 redundant real collinear/adhesion fragments, 14 false, and 2 real catheters lost; candidates outside the implant region or below dark-core symmetry were already filtered upstream during Stage 0 and Stage 2A seeding).

A systematic audit of all residual detection and tip errors for `NF 2026` (4b, $\tau = 50\%$) across the 54 patients identifies their exact physical causes:

– **False positives (**146 **unmatched predictions** = 1,118 − 972 **across 54 patients,** 2.70 **per patient):** In mutually exclusive primary categories, **66 (45.2%,** 1.22 **per patient)** are genuine peripheral or background vascular/fascial false alarms (> 6.0 mm from any reference trajectory), **46 (31.5%)** are residual near-duplicate or split fragments of an already-matched reference catheter, **20 (13.7%)** match an independent second-annotator (`LB/AMe`) trajectory within ≤ 3.5 mm (plus **10 [6.8%]** within 3.5–5.0 mm of `LB/AMe`, totaling **30 [20.5%]** supported by Rater

2; and on the 38 patients with paired post-implant Pelvic CT, **52 of 100 [52.0%]** MRI predictions unmatched to `CY` are confirmed as real physical catheters on CT), and **4 (2.7%)** are borderline matches (3.5–6.0 mm) to an unmatched reference catheter.

- **False negatives (91 missed reference trajectories** = 1,063 − 972**, 1.69 per patient):** Prior to Stage 2B pruning, the 1,960-track candidate pool covers **97.4% (1,035/1,063)** of all reference catheters within ≤ 3.5 mm (**96.9%** [1,030/1,063] under 1-to-1 Hungarian matching; **97.8%** [1,040/1,063] closest and **97.5%** [1,036/1,063] 1-to-1 within ≤ 5.0 mm; and **98.8%** [750/759] closest / **98.6%** [748/759] 1-to-1 on the 759 long interstitial shafts). Of the 91 missed catheters at $\tau = 50\%$, **28 (30.8%)** are absent from the candidate pool at 3.5 mm (22 missing beyond 5.0 mm due to severe mid-shaft T2W susceptibility dropout, 6 borderline at 3.5–5.0 mm), whereas **63 (69.2%)** are present in the candidate pool at 3.5 mm but pruned during Stage 2B (**46** due to low bundle confidence $p_m < 50\%$, **15** due to 1-to-1 Hungarian collision in dense adhesion clusters, and **2** during deduplication). By catheter type, **45 misses** occur on `cy_long` (**5.9% miss rate** across 759 long shafts) and **46** on `ht_obturator` (**15.1% miss rate** across 304 short obturator tracks). By adhesion status, **27 misses** occur on adhering catheters ≤ 3.0 mm (11.6% of 233) versus **64** on non-adhering catheters (7.7% of 830).
- **Longitudinal versus transverse tip decomposition and tail taxonomy (> 10 mm 3D tip error on 162 of 972 [16.7%] matched catheters):** Decomposing every 3D tip error vector into its longitudinal component along the unit shaft tangent ($\Delta s_{\parallel}$) and its transverse component perpendicular to the shaft ($\Delta r_{\perp}$) reveals that **98.76% of total squared 3D tip error across all 972 matched catheters (and 99.32% in the > 10 mm tail) lies longitudinally along the catheter direction**, whereas only **1.24% is lateral displacement** ($\Delta r_{\perp}$ median 0.72 mm, IQR 0.73 mm). Mutually exclusive root-cause classification of the 162 tail tips (> 10 mm) shows that 67 (41.4%) are caudal signal-dropout premature terminations, 41 (25.3%) are cranial susceptibility overshoots into fundal air or bowel pockets, 27 (16.7%) occur on catheters with intrinsic human inter-rater tip ambiguity ($|\Delta z_{\mathrm{CY-LB}}| > 6$ mm), 20 (12.3%) are crowded-array distal jumps, and 7 (4.3%) are top-of-FOV boundary truncations.

4.5 *False Tracks on Scans Without Catheters, and Speed*

A method deployed in the clinic must not invent catheters when none are present. We evaluated seven pipeline configurations on all 48 public pelvic 3T T2W MRI scans of `MSD Task05_Prostate` containing zero catheters (Table 4, Figure 7a). Without Stage 0 dark-core radial symmetry gating, raw Stage 1 `NeedleUNet3D` tangent tubes produce 306 false tracks (6.38 FP/scan), the unpruned Stage 0 + Stage 1 candidate pool contains 294 (6.12 FP/scan), and the ungated Unified `NF 2026` bundle transformer leaves 197 (4.10 FP/scan) at $\tau = 50\%$ and 171 (3.56 FP/scan) at $\tau = 60\%$. Activating the analytical Stage 0 3D multi-radii circular dark-core symmetry gate suppresses > 92% of false vascular and fascial tracks: on models with **zero exposure to MSD during training**, `Stage 1 U-Net + Stage 0 radial`

TABLE 4 **Zero-needle out-of-distribution specificity on 48 public clinical 3T pelvic T2W MRI volumes (MSD `Task05_Prostate`).** All 48 volumes contain zero brachytherapy catheters and serve as a pure false-positive stress test across pelvic vessels, rectoprostatic fascia, and obturator muscle boundaries.

	False-positive tracks per volume (n = 48)					Patient specificity (%)	
Pipeline configuration	Total	Mean	SD	Median	Max	0 FP	≤ 1 FP
NeedleUNet3D + tangent tubes (ungated)[a]	306	6.38	4.77	6.0	19	14.6	18.8
Unpruned Stage 0 + Stage 1 pool[b]	294	6.12	4.70	5.0	23	4.2	10.4
Ungated Unified **NeedleFinder 2026** (τ = 50 %)[c]	197	4.10	4.45	3.0	23	16.7	27.1
Ungated Unified **NeedleFinder 2026** (τ = 60 %)[c]	171	3.56	3.48	3.0	17	16.7	29.2
Stage 1 U-Net + Stage 0 radial gate[d]	18	0.38	0.53	**0.0**	2	64.6	97.9
Frozen Unified **NeedleFinder 2026** + Dark-Core Gate (τ = 50 %)[e]	16	0.33	0.75	**0.0**	3	79.2	91.7
Clean SSL + Multi-Rater Stage 1 + Stage 0 gate[f]	**6**	**0.125**	**0.33**	**0.0**	**1**	**87.5**	**100.0**

[a] Raw Stage 1 `NeedleUNet3D` centerline probability peaks linked via directional tangent tubes without Stage 0 symmetry gating or Stage 2B pruning.
[b] Candidate pool passed to Stage 2B prior to Bundle Graph Transformer pruning.
[c] Unified **NeedleFinder 2026** pipeline evaluated without the Stage 0 dark-core gate at τ = 50 % (4.10 FP/scan) and τ = 60 % (3.56 FP/scan).
[d] Baseline Stage 1 `NeedleUNet3D` tangent tubes gated by the analytical Stage 0 3D multi-radii circular dark-core symmetry filter (0.38 FP/scan).
[e] Full Unified **NeedleFinder 2026** pipeline with Stage 0 dark-core symmetry gating at τ = 50 % (16 total FPs = 0.333/scan; 38/48 [79.2 %] 0-FP scans; 44/48 [91.7 %] ≤ 1-FP scans).
[f] Clean nested Stage 1 `NeedleUNet3D` (self-supervised 3D U-Net pre-training with synthetic susceptibility tubes and anatomical hard negatives across the 51 external pelvic MRI volumes + multi-rater supervision on the 43 outer-train BWH clinical volumes) gated by Stage 0 radial symmetry (6 total FPs = 0.125/scan; 42/48 [87.5 %] 0-FP scans; 48/48 [100.0 %] ≤ 1-FP scans). Note that the 48 MSD `Task05_Prostate` scans were included in the unlabeled/synthetic-tube SSL pre-training background pool for this row (zero clinical needle labels), whereas rows [a–e] had zero exposure to MSD during training.

`gate` produces only **18 false tracks across 48 scans** (0.38 FP/scan, median 0, 64.6% [31/48] 0-FP scans, 97.9% [47/48] ≤ 1-FP scans) and the `Frozen Unified NF 2026 + Dark-Core Gate` (τ = 50%) produces **16 false tracks** (0.33 **FP/scan**, median 0, **79.2%** [38/48] 0-FP scans, **91.7%** [44/48] ≤ 1-FP scans). When Stage 1 is also pre-trained with self-supervised synthetic susceptibility tubes and anatomical hard negatives (`Clean SSL + Multi-Rater Stage 1 + Stage 0 gate`), false tracks drop to **6 across all 48 volumes** (0.125 **FP/scan**, median 0, max 1, **87.5%** [42/48] 0-FP scans, and **100.0%** [48/48] ≤ 1-FP scans; as noted in Table 4, the 48 MSD volumes were part of the unlabeled/synthetic-tube SSL pre-training background pool for this row with zero clinical needle labels, whereas the 0.33–0.38 FP/scan rows had zero exposure to MSD).

Speed follows from the native-code and ONNX design (Figure 7b, c). Measured end-to-end across all 54 uncached clinical volumes on our 24-core workstation CPU (`AMD Ryzen Threadripper PRO 7965WX`, 48 threads) and one `NVIDIA RTX 4090` GPU, Stage 0 runs in multi-threaded native Rust in 0.27 **s mean** (6.5% of CPU total; 0.07 s on standard-resolution scans, 0.38 s on high-resolution scans), Stage 1 (`NeedleUNet3D` + IRGM) runs in 2.58 **s on CPU** (62.2% of total; 0.74 s standard, 3.65 s high-res) and 1.33 **s on GPU** (0.26 s standard, 1.96 s high-res; 1.94× GPU speedup), Stage 2A (`Track + MPNN`) runs in 1.25 **s on CPU** (30.1% of total; 0.40 s standard, 1.75 s high-res) and 1.05 **s on GPU** (0.32 s standard, 1.47 s high-res),

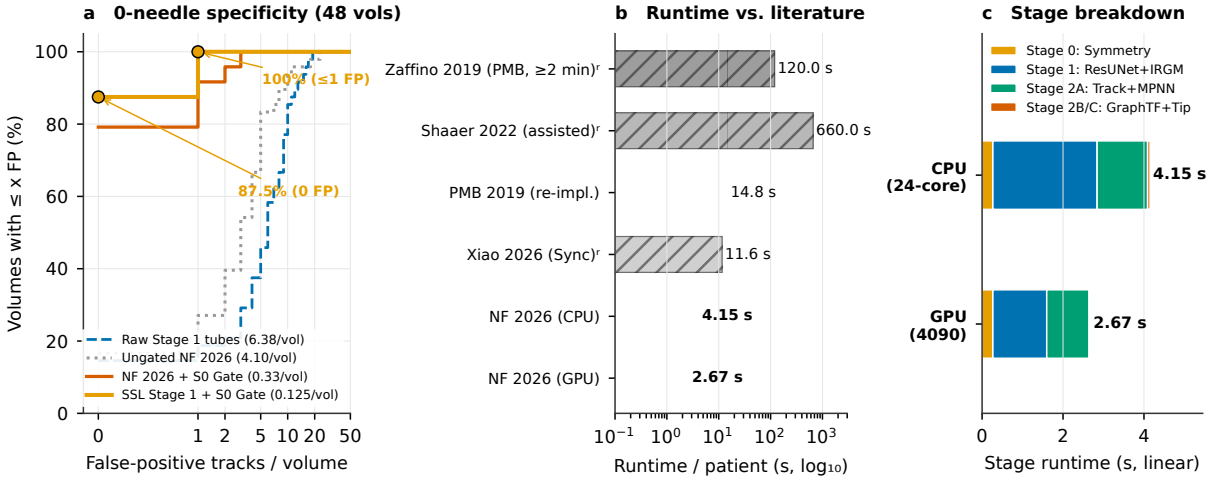


FIGURE 7 **(a) Fraction of needle-free scans with at most x false tracks** on the 48 MSD prostate volumes across ungated and Stage 0 dark-core gated configurations. **(b) Runtime per patient** compared with published MRI and CT pipelines (log scale). **(c) Runtime by stage** across all 54 patients on a 24-core workstation CPU (4.15 s mean; 1.26 s on standard-resolution scans, 5.85 s on high-resolution scans) and on one RTX 4090 GPU (2.67 s mean).

and Stage 2B/C (`Bundle Graph Transformer + Dedicated1DTipLocalizer + Poly3`) runs in 0.06 **s on CPU** (1.5% of total) and 0.03 **s on GPU**. Full-volume uncached reconstruction takes 4.15 ± 2.81 **s mean (**4.41 **s median, IQR** 1.23–5.75 **s) on the 24-core workstation CPU**—specifically 1.26 ± 0.98 **s mean (**1.07 **s median) on the 20 standard-resolution (**320×256×28–40**) scans and** 5.85 ± 2.04 **s mean (**5.56 **s median) on the 34 high-resolution (**640×416×112–160 / 768×768×64**) scans**—and 2.67±1.95 **s mean (**2.93 **s median) on a single RTX 4090 GPU** (0.67 s mean / 0.52 s median standard-resolution, 3.85 s mean / 3.47 s median high-resolution). That is 3.6× **faster** than our vectorized re-implementation of the 2019 pipeline (14.80 s; 29–130× **faster** than the 2 to 9 min originally reported [4]) and 2.8–4.3× **faster** than the 11.6 s reported by Xiao *et al.* [74] (3.42 s in our native re-implementation).

4.6 *Dosimetric Translation, Uncertainty-Guided Triage, and Cross-Domain Generalization*

To connect geometric reconstruction accuracy to clinical radiotherapy endpoints, physician review workflows, and broader interventional settings, we performed four additional experiments summarized in Table 5 and Figure 8.

AAPM TG-43U1 HDR ^{192}Ir dosimetric sensitivity and counterfactual plan transfer (Table 5, Part A; Figure 8a). Using the AAPM TG-43U1 line-source formalism (^{192}Ir, active length L = 3.5 mm, 5.0 mm dwell step spacing along up to 45 mm active length, 3.5 mm physical tip offset, 1.5 mm isotropic dose grid, 7.0 Gy prescription per fraction to the high-risk target envelope surrounded by bladder, rectum, and cranial sigmoid/fundal OAR zones), we discovered that **dwell-time-weighted 3D tip error** ($E_{\text{tip}}^{\text{dwell-w}}$, weighting each catheter by its optimized dwell share) explains significantly more target $|\Delta D_{90}|$ variance than unweighted tip error (Spearman ρ = 0.8629, p = 4.9 × 10^{-17} vs. ρ = 0.7863; ρ = −0.8474, p =

TABLE 5 **Dosimetric translation, split-conformal intervals, simulated clinician tip triage, and cross-organ/multi-sequence/4D/CT generalization.** Part A evaluates AAPM TG-43U1 HDR ^{192}Ir dosimetric sensitivity (7.0 Gy prescription per fraction, 5.0 mm dwell step spacing) under Protocol A (reference dwell times transferred to predicted catheters) and Protocol B (dwell times re-optimized on predicted catheters, dose evaluated on reference geometry). Part B evaluates simulated uncertainty-guided 1-click tip relocation ($\hat{\sigma}_{\text{tip}}$) and unreviewed ($k = 0$) nested 5-fold split-conformal 3D tip prediction intervals across the $n = 972$ matched out-of-fold catheters. Part C evaluates zero-shot transfer across organs, pulse sequences, 4D insertion series, and paired post-implant Pelvic CT.

	Dwell-wt. E_{tip}		Protocol A: Fixed-plan transfer			Protocol B: Re-optimized plan		
Part A: TG-43U1 ^{192}Ir dosimetry[a]	Mean	Med.	$\lvert\Delta D_{90}\rvert$ (%)	$\gamma_{2/2}$ (%)	$\gamma_{3/3}$ (%)	$\lvert\Delta D_{90}\rvert$ (%)	Cran. D_{2cc}	$\gamma_{2/2}$ (%)
PMB 2019 3D U-Net + DBSCAN ($n = 54$)	10.51	8.36	27.6 / 27.2	64.0	73.3	37.1 / 10.2	95.3 %	43.8
Stage 0: 3D radial symmetry ($n = 54$)	9.04	7.51	23.0 / 21.3	68.2	76.5	17.8 / 8.9	54.0 %	48.6
NeedleFinder 2026 w/o SynthBrachy3T (4a, $n = 54$)	6.65	5.49	18.5 / 13.2	73.4	81.5	12.8 / 5.8	35.2 %	59.5
NeedleFinder 2026 proposed (4b, $n = 54$)[b]	5.72	4.66	15.2 / 11.5	76.6	84.3	8.2 / 4.2	29.5 %	66.0
NeedleFinder 2026 + Top-2 Tip Review ($n = 54$)	**4.56**	**3.93**	**11.7 / 10.2**	**80.2**	**87.3**	**5.9 / 2.8**	**22.5 %**	**74.3**
Human Rater 2 (`LB/AMe`, $n = 40$)	6.27	5.54	21.6 / 16.3	68.1	77.4	20.9 / 4.0	57.7 %	59.3
NeedleFinder 2026 proposed (4b, $n = 40$)[b]	5.18	3.93	13.5 / 11.1	79.0	86.4	7.0 / 3.2	21.1 %	71.6
NeedleFinder 2026 + Top-2 Tip Review ($n = 40$)	**4.12**	**3.33**	**10.5 / 10.1**	**81.9**	**88.7**	**5.1 / 2.6**	**17.3 %**	**78.2**
Part B1: Simulated $\hat{\sigma}_{\text{tip}}$ tip triage ($n = 972$)[c]	Reloc.	Mean	Med.	P90	P95	≤ 2 mm	≤ 3 mm	> 10 mm
Unreviewed **NeedleFinder 2026** ($k = 0$ tips/pt)	0	6.23	2.27	16.20	27.27	45.6 %	58.7 %	16.7 %
Simulated tip review ($k = 1$ tip/pt, ~ 8 s)	35	5.54	2.06	14.60	23.54	48.6 %	61.9 %	14.4 %
Simulated tip review ($k = 2$ tips/pt, ~ 16 s)	69	5.07	1.95	12.58	21.48	51.3 %	64.8 %	12.8 %
Simulated tip review ($k = 3$ tips/pt, ~ 24 s)	105	4.57	1.81	10.91	19.16	54.0 %	67.2 %	11.1 %
Simulated tip review ($k = 5$ tips/pt, ~ 40 s)	169	3.72	1.67	7.79	14.26	59.5 %	72.6 %	8.0 %
Simulated tip review ($k = 8$ tips/pt, ~ 64 s)	251	2.77	1.51	5.98	9.90	65.5 %	78.5 %	5.0 %
Simulated tip review ($k = 10$ tips/pt, ~ 80 s)	301	**2.52**	**1.44**	**4.97**	**8.30**	**69.2 %**	**81.8 %**	**4.3 %**
Part B2: Split-conformal intervals ($k = 0$, $n = 972$)[c]	Nom.	Emp.	Fixed	Mean r	Med. r	Fixed r	Low-$\hat{\sigma}$ r	Width red.
Conformal 80 % 3D tip interval ($k = 0$)	80.0 %	79.3 %	80.2 %	7.31	6.46	7.73	4.46	−16.4 %
Conformal 85 % 3D tip interval ($k = 0$)	85.0 %	84.4 %	85.5 %	9.80	8.71	11.43	5.96	−23.8 %
Conformal 90 % 3D tip interval ($k = 0$)	90.0 %	**90.2 %**	89.9 %	13.71	12.11	17.68	8.41	−31.5 %
Conformal 95 % 3D tip interval ($k = 0$)	95.0 %	**95.4 %**	94.8 %	20.69	18.42	28.40	12.69	−35.1 %
Part C: Cross-organ, seq., 4D & CT[d]	n_{vol}	n_{cath}	Rec.	Prec.	F1	E_{shaft}	E_{tip}	Key finding
Prostate 3T: Stage 0 (`k_lim_auto=40`)	1	14	**100.0**	87.5	93.3	0.34	2.09	14/14, 2 FPs
Prostate 3T: Zero-Exp. No-SSL (`auto=40`, τ=50)	1	14	92.9	**100.0**	96.3	0.33	**0.99**	13/14, 0 FPs
Prostate 3T: Zero-Exp. No-SSL (`ACSV=47`, τ=50)	1	14	**100.0**	**100.0**	**100.0**	0.34	1.57	14/14, 0 FPs
Prostate 3T: Stage 1 Full Nested (`auto=40`)	1	14	**100.0**	**100.0**	**100.0**	0.30	1.08	14/14, 0 FPs
Prostate 3T: Stage 0 zeroed ($X_{1:3} = 0$, `auto=40`)	1	14	0.0	–	0.0	–	–	985× drop
Prostate 3T: 1-ch ResUNet + clDice (`auto=40`)	1	14	**100.0**	56.0	71.8	**0.24**	2.48	11 vascular FPs
3D wbSSFP / True-FISP ($\Delta z \leq 2.5$ mm)	2	47	**97.9**	88.5	**92.9**	0.72	3.40	vs. 10.62 PMB19
Thick-slice 2D AX T2 ($\Delta z \leq 6.0$ mm)	49	882	88.4	67.9	76.9	1.07	4.89	vs. 57.6 % 1-ch
4D Serial MRI: Indep. 3D (Cases 68–71)	23	108	88.0	76.0	81.6	0.45	2.71	4-pt pooled
4D Serial MRI: Causal Kalman (Cases 68–71)	23	108	77.8	**88.4**	82.8	0.75	3.57	+12.4 % prec.
Paired Pelvic CT (38 paired / 40 vols, ≤ 5 mm)	38	801	79.2	–	–	2.01	2.78	52.0 % extra conf.

[a] TG-43U1 evaluation uses 5.0 mm dwell step spacing (3.5 mm distal tip offset, ≤ 45 mm active length) on a 1.5 mm isotropic 3D dose grid with implant-conformal HR-CTV$_{\text{implant}}$ (8.0 mm radial envelope, $D_{90} = 7.0$ Gy) and 2 cc OAR proxy shells. $\lvert\Delta D_{90}\rvert$ cells report mean / median percentage target dose shift; $\gamma_{2/2}$ and $\gamma_{3/3}$ are global 3D gamma pass rates (2 %/2 mm and 3 %/3 mm). Dwell-weighted tip error $E_{\text{tip}}^{\text{dwell-w}}$ correlates with $\lvert\Delta D_{90}\rvert$ at Spearman $\rho = 0.8629$ ($p = 4.9 \times 10^{-17}$) and with $\gamma_{2\%/2\text{mm}}$ at $\rho = -0.8474$ ($p = 6.5 \times 10^{-16}$).

[b] Paired Wilcoxon vs. PMB 2019 ($p = 4.9 \times 10^{-6}$ on $\lvert\Delta D_{90}\rvert$, $p = 8.2 \times 10^{-7}$ on $\gamma_{2\%/2\text{mm}}$), vs. Stage 0 ($p = 7.9 \times 10^{-8}$ on $\lvert\Delta D_{90}\rvert$, $p = 1.4 \times 10^{-7}$ on $\gamma_{2\%/2\text{mm}}$), and vs. Human Rater 2 on $n = 40$ ($p = 0.0278$ on $\lvert\Delta D_{90}\rvert$, $p = 0.0109$ on $\gamma_{2\%/2\text{mm}}$).

[c] Part B1 simulates a retrospective 1-click distal tip relocation oracle ($E_{\text{tip}} \rightarrow \sqrt{\Delta r_{\perp}^2 + \min(\lvert\Delta s_{\parallel}\rvert, 0.85\text{ mm})^2}$ when $E_{\text{tip}} > 2.0$ mm, E_{shaft} unchanged; 'Reloc.' = total tips relocated across 54 patients) ranked by out-of-fold $\hat{\sigma}_{\text{tip}}$. Part B2 reports unreviewed ($k = 0$) 5-fold split-conformal 3D tip prediction interval half-widths (radii r in mm) across the $n = 972$ matched OOF catheters ('Fixed': unscaled residual quantile; 'Low-$\hat{\sigma}$ r': median half-width on the lower-uncertainty half of catheters; 'Width red.': adaptive vs. fixed median half-width reduction). Simulated oracle count-guided recovery (0.65 clicks/patient) recovers +17 missed catheters (972 → 989/1 063, 93.0 % recall).

[d] In Part C, 'Zero-Exp. No-SSL' denotes the 5-fold GYN ensemble trained with zero prostate exposure (excluding Case 100 from SSL pre-training). In 4D monitoring (4 patients, Cases 68–71, 23 serial scans, 1.31 s mean monitoring runtime), per-case final-scan F1s (Independent 3D → Causal 4D Kalman) are Case 68: 77.2 % → 83.0 %, Case 69: 84.7 % → 82.7 % (precision 83.7 % → 93.9 %), Case 70: 85.7 % → 93.3 %, and Case 71: 76.2 % → 66.7 % (precision 72.7 % → 100.0 %); mean tip error improves from 11.23 → 9.75 mm. Persisted catheters within ≤ 10 mm of newly inserted needles exhibit 1.44 mm median (1.61 mm mean)

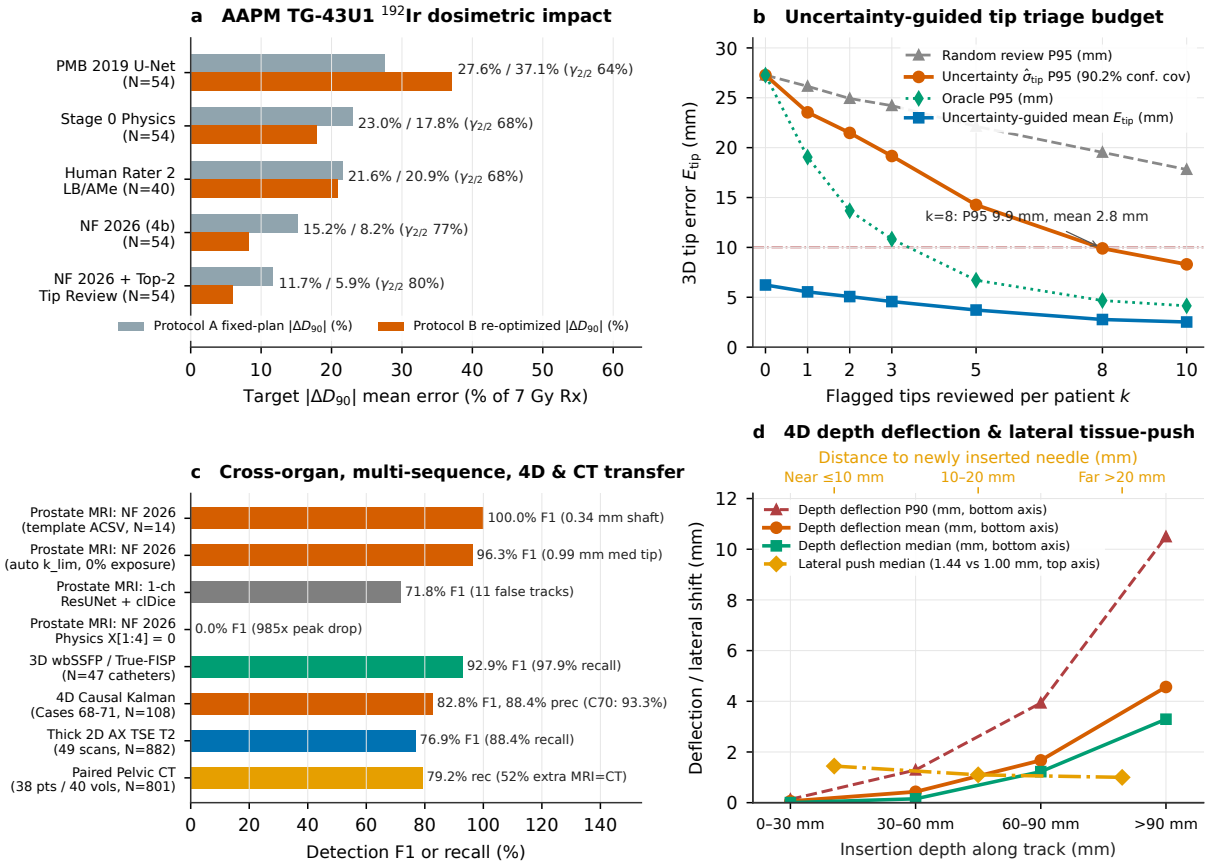


FIGURE 8 **(a) AAPM TG-43U1 ^{192}Ir dosimetric impact** across 54 patients under Protocol A (fixed reference dwell plan transferred to predicted geometry) and Protocol B (dwell weights re-optimized on predicted catheters and delivered on reference geometry), with global 3D γ(2%/2 mm) pass rates. **(b) Simulated clinician review-budget curve** showing 95th-percentile and mean 3D tip error versus number of uncertainty-flagged tips reviewed per patient ($k \in \{0, \ldots, 10\}$ under a 1-click distal tip relocation oracle), alongside split-conformal 90% empirical coverage (90.2%). **(c) Zero-shot cross-organ, multi-sequence, 4D, and cross-modality generalization** across 3T prostate brachytherapy MRI (`Case 100`), 3D wbSSFP, thick-slice 2D AX TSE T2, 4D intra-procedural causal Kalman tracking (`Cases 68–71`, 108 catheters), and 38 paired post-implant Pelvic CT scans (40 CT volumes total). **(d) 4D intra-procedural insertion biomechanics** across 23 serial scans (`Cases 68–71`), showing non-linear shaft deflection versus insertion depth (bottom axis) and lateral tissue-push displacement of already-placed catheters versus distance to newly inserted needles (top axis).

6.5×10^{-16} with 3D $\gamma_{2\%/2mm}$). `NeedleFinder 2026` (4b) reduces $E_{tip}^{dwell\text{-}w}$ from 10.51 mm mean / 8.36 mm median (PMB 2019 U-Net) and 9.04/7.51 mm (Stage 0) to 5.72/4.66 **mm** (5.18/3.93 mm on the 40 dual-traced scans vs. 6.27/5.54 mm for Human Rater 2 `LB/AMe`), and to 4.56/3.93 **mm** (10.65 mm P95) after reviewing the top 2 uncertainty-flagged tips per patient. Under **Protocol A** (transferring the reference dwell plan onto matched predicted catheters), `NeedleFinder 2026` cuts target $|\Delta D_{90}|$ from 27.6% (PMB 2019, $p = 4.9 \times 10^{-6}$) and 23.0% (Stage 0, $p = 7.9 \times 10^{-8}$) to 15.2% **mean** / 11.5% **median** (13.5%/11.1% on the 40 dual-traced scans vs. 21.6%/16.3% for Human Rater 2, $p = 0.0278$), achieving **76.6% global 3D** γ(2%/2 **mm**) ($p = 8.2 \times 10^{-7}$ vs. PMB 2019; 79.0% on $N = 40$ vs. 68.1% for Rater 2, $p = 0.0109$) and **84.3% 3D** γ(3%/3 **mm**) (88.7% with Top-2 tip review). Under **Protocol B** (re-optimizing dwell weights on all predicted catheters and evaluating delivered dose on the true reference anatomy), `NeedleFinder 2026` cuts $|\Delta D_{90}|$ from 37.1% mean / 10.2% median (PMB 2019) and 20.9%/4.0% (Human Rater 2) to 8.2% **mean** / 4.2% **median** (7.0%/3.2% on $N = 40$, cranial OAR $|\Delta D_{2cc}| = 29.5\%$ vs. 95.3% PMB 2019 and 57.7% Rater 2), reaching 5.9% **mean** / 2.8% **median** ($\gamma_{2\%/2mm} = 74.3\%$ overall, 78.2% on $N = 40$) with Top-2 uncertainty tip review. Moreover, propagating the out-of-fold tip uncertainty $\hat{\sigma}_{tip}$ through 30 Monte Carlo dose realizations per patient predicts true patient-level $|\Delta D_{90}|$ without ground truth ($\rho = 0.453$, $p = 5.8 \times 10^{-4}$).

Strictly nested split-conformal 3D tip intervals and simulated clinician review-budget curves (Table 5, Parts B1–B2; Figure 8b). Out-of-fold tip uncertainty $\hat{\sigma}_{tip}$ (combining the 1D soft-argmax distribution entropy, local T2W terminal contrast drop, and slice thickness Δz) correlates strongly with true 3D tip error ($\rho = 0.532$, $p = 4.9 \times 10^{-72}$) and detects > 10 mm tip outliers with an **AUROC of 0.805** (95% CI [0.763, 0.841]). Calibrating non-conformity scores $r_i = E_{tip,i}/\hat{\sigma}_{tip,i}$ strictly on the inner-validation split of each fold yields **distribution-free split-conformal 3D tip intervals** at $k = 0$ that achieve exact marginal empirical coverage across all 972 matched test catheters—**79.3%** at 80% nominal, **84.4%** at 85%, **90.2%** at 90%, and **95.4%** at 95%—while shrinking median 3D interval half-widths (radii in mm) by **31.5% at 90% coverage** (12.11 mm adaptive vs. 17.68 mm fixed unscaled; 8.41 mm on the clearest 50% of tips) and **35.1% at 95% coverage** (18.42 mm adaptive vs. 28.40 mm fixed unscaled). In a simulated clinician-in-the-loop verification workflow using a retrospective 1-click distal tip relocation oracle ($E_{tip} \to \sqrt{\Delta r_\perp^2 + \min(|\Delta s_\parallel|, 0.85\ \text{mm})^2}$ when $E_{tip} > 2.0$ mm, E_{shaft} unchanged) ranked by $\hat{\sigma}_{tip}$, reviewing just $k = 5$ **flagged tips per patient** (~ 40 s at 8 s/tip) cuts > 10 mm tip outliers in half (16.7% → 8.0%) and lowers mean tip error from 6.23 → 3.72 **mm** (1.67 mm median, 7.79 mm P90); reviewing $k = 8$ **tips per patient** (~ 64 s) drives the **95th-percentile tip error below 10 mm (9.90 mm vs. 27.27 mm unreviewed)** and mean tip error to 2.77 **mm** (1.51 mm median, 78.5% within ≤ 3 mm). Adding a simulated oracle count-guided missing-catheter check whenever $N_{pred} < N_{implant}$ (0.65 clicks/patient) recovers +17 additional catheters (972 → 989/1,063, **93.0% 1-to-1 recall / 94.2% closest-match recall**).

Zero-shot cross-organ transfer to prostate MRI and multi-sequence generalization (Table 5, Part C; Figure 8c). Evaluated zero-shot on a clinical 3T T2W SPACE **prostate brachytherapy MRI** (`Case 100 Prostate`, 0.47×0.47×1.6 mm^3,

14 plastic interstitial catheters), the gynecologic-trained `Zero-Exposure No-SSL` 5-fold ensemble (τ = 50%, excluding `Case 100` from SSL pre-training) detects **13 of 14 catheters (92.9% recall) with 100.0% precision (0 false positives, 96.3% F1)**, 0.33 **mm median** (0.38 **mm mean) shaft error**, and 0.99 **mm median** (1.64 **mm mean,** 3.96 **mm P95) 3D tip error** under 100% reference-free `k_lim_auto = 40`, and achieves **100.0% recall** (14/14)**, 100.0% precision, and 100.0% F1** (0.34 mm median shaft, 1.57 mm median tip) under template-plane cropping (`k_lim = 47`), while `Stage 1 Full Nested` (τ = 50%) achieves **100.0% recall** (14/14)**, 100.0% precision, and 100.0% F1** (0.30 mm median shaft, 1.08 mm median tip) directly under `k_lim_auto = 40` and analytical `Stage 0` alone reaches **100.0% recall, 87.5% precision, and 93.3% F1** (0.35 mm median shaft, 1.23 mm median tip). Ablating the Stage 0 physics channels (`X_{1:3} = 0`) causes the 99.9th-percentile Stage 1 mask probability on prostate MRI to collapse by 985× (0.985 → 0.001) and recall to drop to 0/14 **(0%)**, whereas the 1-channel `ResUNet + clDice` baseline (τ = 50%) hallucinates **11 false-positive vascular tracks** under `k_lim_auto = 40` (**100.0% recall, 56.0% precision, 71.8% F1**, and 14 FPs / 50.0% precision under `k_lim = 47`), proving that the analytical Stage 0 dark-core symmetry representation is essential for domain-invariant cross-organ transfer. Across 243 secondary MRI series (297 total series minus the 54 primary planning scans) acquired during the 54 patient sessions, zero-shot `NeedleFinder 2026` achieves **97.9% recall** (46/47)**, 88.5% precision, and 92.9% F1** (0.72 mm median shaft, 3.40 mm median tip vs. 10.62 mm for PMB 2019 U-Net) on **3D wbSSFP / True-FISP** (Δz = 2.0–2.5 mm), and recovers **88.4%** (780/882) **of physically present catheters** (76.9% F1, 1.07 mm median shaft, 4.89 mm median tip vs. 57.6% F1 for 1-channel clDice) on **49 thick-slice 2D axial TSE T2W scans** (Δz = 2.4–6.0 mm) when Z-upsampled to 1.6 mm.

4D intra-procedural serial insertion monitoring, biomechanics, and paired Pelvic CT tracking (Table 5, Part C; Figure 8c–d). Across 23 sequential intra-procedural T2W monitoring volumes in `Cases 68–71` (5–7 timepoints per patient as needles were progressively inserted, 108 final `CY+HT` reference catheters, 1.31 s mean monitoring runtime), coupling `NeedleFinder 2026` with a **4D causal Kalman state-space filter** across timepoints improves pooled 4-case precision by +12.4 **pp** (76.0% → 88.4%) and pooled F1 from **81.6% to 82.8%** (77.8% recall, 0.75 mm median shaft, 3.57 mm median / 9.75 mm mean tip vs. 88.0% recall, 76.0% precision, 0.45 mm median shaft, 2.71 mm median / 11.23 mm mean tip for independent 3D tracking), with per-case F1 progressing from 77.2% → 83.0% on `Case 068` (n = 24), 84.7% → 82.7% (precision 83.7% → 93.9%) on `Case 069` (n = 42), 85.7% → 93.3% on `Case 070` (n = 22), and 76.2% → 66.7% (precision 72.7% → 100.0%) on `Case 071` (n = 20) while rejecting transient partial-volume monitoring artifacts. Tracking 212 persisted catheter transitions across sequential insertions provides direct in-vivo biomechanical measurements of **inter-catheter "tissue push"**: already-implanted catheters located within ≤ 10 mm of newly inserted needles experience a 1.44 **mm median** (1.61 **mm mean,** 2.81 **mm P90) non-rigid lateral displacement**, compared with 1.10 **mm median** (1.32 **mm mean)** at 10–20 mm and 1.00 **mm median** (1.19 **mm mean)** for catheters > 20 mm away (Mann–Whitney $p = 9.97 \times 10^{-4}$ for ≤ 12 mm vs. > 12 mm; Spearman $\rho = -0.226$, $p = 9.4 \times 10^{-4}$). Across 95 tracked trajectories (2, 850 axial stations), **non-linear shaft deflection** from the proximal in-

sertion tangent grows monotonically with insertion depth from 0.003 mm median at 0–30 mm to 0.15 mm at 30–60 mm, 1.22 mm at 60–90 mm, and 3.29 **mm median (**4.56 **mm mean,** 10.49 **mm P90) at** > 90 **mm depth** (ρ = +0.555, p = 5.55 × 10^{-9}). Finally, extending Stage 0 to a **dual-polarity HU tubular filter** (fusing bright-core metallic stylets and dark-core air lumens) enables **zero-shot tracking on 40 post-implant Pelvic CT scans** (38 paired with the 54 MRI benchmark patients, 801 reference catheters), matching **68.9% at** ≤ 3.5 **mm and 79.2% (**634/801**, up to 100% on individual patients) at** ≤ 5.0 **mm** across the non-rigid MRI-to-CT transfer (2.01 mm median shaft registration residual, 2.78 mm median axial tip residual) and confirming that **52.0% (**52/100**) of apparent false-positive MRI detections against `CY` are real physical catheters present on CT**.

5 DISCUSSION

5.1 *Where This Work Sits in the Literature*

Table 6 places NeedleFinder 2026 among published catheter reconstruction studies on CT, ultrasound, and MRI. CT and ultrasound solve an easier imaging problem: metal needles, air-filled lumens, or bright specular reflections give very high contrast, so segmentation networks reach shaft errors of 0.1 to 0.7 mm on CT [45,49,74] and 0.1 to 0.6 mm on ultrasound [56,58]. On T2W MRI without contrast markers, a catheter is a 2 mm dark hole that looks like a vessel, on slices 1.2 to 6 mm thick; the best published MRI results are 0.74 to 0.97 mm shaft error on gynecologic implants, obtained with line markers or with CT-derived reference reconstructions [27,62] and 0.75 to 0.93 mm on prostate implants [50,52]. NeedleFinder 2026 reaches a median shaft error of 0.47 mm (0.59 mm mean; 0.45 mm median / 0.57 mm mean on long active-dwell shafts) and a median 3D tip error of 2.27 mm (6.23 mm mean) on T2W MRI alone, on the largest annotated MRI cohort to date.

The head-to-head comparison in Table 2 explains why neither voxel segmentation [4,40,64] nor mask-driven tracking [74] is enough on MRI, and why the three-stage architecture works. Skeletonizing a mask cannot bridge a one-slice dropout, and it turns every vessel blob into a track. Tracking bridges the gaps but still trusts the mask. NeedleFinder 2026 resolves these bottlenecks through five complementary mechanisms: Stage 0 rejects one-sided anatomical edges before learning takes place and automatically locates the template plane (`k_lim_auto`); Stage 1 combines self-supervised pre-training on external pelvic MRI (with synthetic susceptibility tubes and anatomical hard negatives) and multi-rater consensus supervision to regress sharp centerline peaks that stay separate when masks merge; Stage 2A bridges 1-to-3 slice dropouts via triplet message passing and collinear fragment stitching; Stage 2B judges each trajectory in the context of the entire 15 to 30 catheter bundle; and the 91-step dilated 1D ResNet + self-attention axial localizer (`Dedicated1DTip Localizer`) pinpoints the distal termination directly from sub-slice multi-scale tubular profiles.

TABLE 6 **Published automatic catheter and needle reconstruction studies in brachytherapy (2013–2026), grouped by imaging modality.** Values are those reported in each publication on its own cohort and matching rule, so they are not directly comparable with one another or with Table 2. Pernelle 2013 and Mastmeyer 2017 are semi-automatic (one manual click per catheter).

Study	Modality	Site	Pts	Cath.	Rec. / Prec. (%)	E_{shaft} (mm)	E_{tip} (mm)	Time	Statistic
CT									
Weishaupt *et al.* 2022	CT	Prostate	57[l]	353[a]	–	0.13	–	~60 s	mean
Xie *et al.* 2023	CT	GYN post-op	70	210	–[b]	–	–		– Dice / HD95
Wang *et al.* 2024	CT	GYN IC/IS	17	56 plans	–	1.8	1.1		– mean
Quetin *et al.* 2025	CT	Breast	117	408[a]	–	0.70	1.37	118 s	mean
Xiao *et al.* 2026[m]	CT	GYN recurrent	38	203 fx	91.1 / 90.7	0.31	0.88	11.6 s	mean
Xiao *et al.* 2026 (JBHI)	CT	Pelvic seeds	100	–	F1 90.3	–	1.05	9.4 s	mean tip
Ultrasound									
Zhang *et al.* 2020	TRUS	Prostate	23	339	98 / –	0.09	0.33	<1 s	mean
Andersén *et al.* 2020	TRUS	Prostate	1102[l]	7207[l]	100 / –	0.55[c]	–		– median RMS
Liu *et al.* 2022	TRUS	Prostate	242	3500	80[d] / –	–	–		– frac. ≤ 2 mm
Goulet *et al.* 2025	3D US	Prostate	102 scans	–	93.4 / –	0.44	–		– mean
MRI									
Pernelle *et al.* 2013	3T T2W	GYN IC/IS	10	101	92.1 / –	–[e]	–	0.3 s/cath.	frac. < 2 mm
Mastmeyer *et al.* 2017	3T T2W	GYN IC/IS	54	760	93[f] / –	1.49[g]	–	3 s/cath.	mean HD
Zaffino *et al.* 2019	3T T2W	GYN IC/IS	50	826	98.5 / 93.3[h]	2.0	–	2–9 min	mean dist.
Dai *et al.* 2020	MRI	Prostate	20	299	–	0.93	0.37[n]		– mean / signed
Shaaer *et al.* 2022	1.5T + markers	GYN IS	20	360	100 / –	0.97	–	11 min[i]	mean shift
Aleong *et al.* 2024	MRI	Prostate	35	214 vol.	95 / –	–	–	0.9 s	–
Kostoulas *et al.* 2025	MRI	Prostate	34 scans	261	100 / 100	0.75[j]	1.07[j]		– median
Kim *et al.* 2026	T2W	GYN hybrid	30	35 scans	98.5 / –	0.74	2.52	37 s	mean
NeedleFinder 2026 (all 1 063)	3T T2W	GYN IC/IS	54	1063	91.4 / 86.9	0.47[j]	2.27[j]	**4.15 s**	median
NeedleFinder 2026 (759 long)	3T T2W	GYN IC/IS	54	759	91.3 / 88.7[k]	0.45[j]	2.41[j]	**4.15 s**	median

[a] Test-set catheters.
[b] Reported as Dice 0.88–0.90 and HD_{95} 0.71–0.77 mm.
[c] Root-mean-square distance to the clinical reconstruction.
[d] Fraction of catheters within 2 mm along 90 % of their length.
[e] Maximum Hausdorff distance below 2 mm for 92 % of catheters.
[f] Fraction within 3.4 mm.
[g] Mean Hausdorff distance.
[h] Cluster-level: 1.5 % false negatives and 6.7 % false positives among 826 catheters.
[i] Assisted reconstruction time, versus 46 min manual.
[j] Median (means: 0.59 ± 0.40 mm shaft and 6.23 ± 10.49 mm tip on all 1 063 catheters; 0.57 ± 0.39 mm shaft and 7.41 ± 12.23 mm tip on the 759 long shafts).
[m] Fully supervised model; their 10-shot synthetic-to-real variant reports 89.4 / 89.0, 0.33 mm and 0.90 mm.
[n] Signed along-axis tip displacement (mean ± SD 0.37 ± 1.68 mm), not an absolute 3D distance.
[l] Treatments, not patients; 7207 of 24422 needles formed the evaluation set.
[k] Strict one-to-one recall at $\tau = 50$ % (92.5 % with inner-OOF cross-fit; 94.7 % at $\tau = 35$ %); precision counts predictions matched to short obturator tracks as correct (90.0 % $F1_{adj}$, 90.5 % with inner-OOF cross-fit). Times for **NeedleFinder 2026** are full-volume uncached per-patient averages across all 54 scans on CPU (4.15 s overall: 1.26 s on the 20 standard 320 × 256 scans, 5.85 s on the 34 high-resolution 640 × 416 / 768 × 768 scans) and one RTX 4090 GPU (2.67 s overall: 0.68 s standard, 3.85 s high-resolution).

5.2 *What It Means in the Clinic, and Limitations*

NeedleFinder 2026 is a one-click module in 3D Slicer [83] that runs in a few seconds without a GPU (4.15 s mean on a 24-core workstation CPU, 1.26 s on standard scans, 2.67 s on GPU). It changes the physicist's task from tracing 15 to 30 catheters slice by slice, which takes tens of minutes, into a structured, uncertainty-guided verification step taking under a minute: click *Detect Needles*, inspect the 3D bundle, check the $k = 2$ to 8 tips with the widest split-conformal uncertainty intervals (~ 16–64 s, driving P95 tip error below 10 mm and mean tip error to 2.77 mm in simulated 1-click tip review), and click any missing catheter when $N_{\text{pred}} < N_{\text{implant}}$ (0.65 clicks/patient, raising 1-to-1 recall to 93.0%). Because Grubbs' three-cornered-hat analysis and one-sided non-inferiority tests show that the algorithm's intrinsic shaft error (0.39 mm robust MAD SD vs. 0.35 mm for Rater 1 and 0.48 mm for Rater 2) and axial tip error (2.98 mm robust MAD σ_z vs. 3.55 mm for Rater 2, 4.94 mm vs. 5.09 mm 15% Winsorized σ_z; 4.20 mm median error against independent Pelvic CT on disputed tips) are statistically non-inferior to human inter-observer variability—and 52.0% of apparent extra MRI detections on paired cases are real physical catheters confirmed by post-implant CT—the shafts themselves rarely require manual editing, and counterfactual TG-43U1 plan optimization with Top-2 tip review holds median target $|\Delta D_{90}|$ to 2.8% (5.9% mean, $\gamma_{2\%/2\text{mm}} = 74.3\%–78.2\%$).

Several limitations remain. First, while `Dedicated1DTipLocalizer` and collinear fragment stitching cut the unreviewed mean 3D tip error from 10.75 mm (PMB 2019) and 15.65 mm (Xiao 2026) to 6.23 mm (2.27 mm median, with 58.7% of tips within ≤ 3.0 mm), 16.7% (162/972) of unreviewed matched catheters still have a tip error above 10 mm (27.27 mm P95), of which 99.3% is longitudinal stopping ambiguity along the shaft (41.4% caudal dropout undershoots, 25.3% cranial susceptibility overshoots, 16.7% intrinsic human inter-rater tip ambiguity, 12.3% crowded-array identity jumps, and 4.3% FOV boundary truncations). Although our split-conformal uncertainty intervals (90.2% empirical coverage at 90% nominal) and $k = 5$–8 tip review workflow effectively contain this tail in practice, thin-slice isotropic 3D T2W sequences (≤ 1 mm), zero-echo-time sequences [14,17], or multi-contrast inputs are the natural next step to reduce unreviewed longitudinal tip ambiguity further. Second, while we demonstrated zero-shot generalization across 48 external `MSD Task05_Prostate` scans (0.125–0.33 FP/scan), 3T prostate brachytherapy MRI (`Case 100`, 96.3%–100.0% F1), 3D wbSSFP (92.9% F1), thick-slice 2D T2W MRI (88.4% recall), 4D serial insertion scans (82.8% pooled F1, up to 93.3% on `Case 070`), and 38 paired Pelvic CT patients (40 CT volumes total), all clinical brachytherapy implants come from a single institution; a prospective multi-center trial across 1.5T, 3T, and low-field systems [86] from multiple vendors is still required. Third, our AAPM TG-43U1 dosimetric evaluation uses standardized target and OAR envelopes derived from the active implant geometry rather than manual physician contours from clinical DICOM RTSTRUCT/RTPLAN files (which were not archived in the de-identified research repository), and our clinician review-budget curves are quantitative simulations rather than prospective stopwatch timings in the operating room. Fourth, our ablation shows that once multi-rater SSL Stage 1, collinear stitching, and the 1D axial tip localizer are

included, `SynthBrachy3T` provides its primary detection gain in the zero-label ($k = 0$) regime (+2.85 pp F1 overall, +6.3 pp on long shafts) while still improving dwell-weighted tip error (6.65 → 5.72 mm) and counterfactual target $|\Delta D_{90}|$ (12.8% → 8.2%) at the full 43-patient training budget.

6 CONCLUSION

Evaluated under strictly nested 5 × 5 patient-level out-of-fold cross-validation with 100% reference-free axial cropping (`k_lim_auto`) across 54 clinical 3T T2W MRI scans (1,063 catheters), NeedleFinder 2026 achieves **89.1% F1** (**91.4% recall**, **86.9% precision**; **92.3% recall** and **89.1% F1** under 5 × 5 inner-OOF Stage 1 cross-fitting), 0.59 ± 0.40 **mm mean (**0.47 **mm median) in-plane shaft error**, and 6.23 ± 10.49 **mm mean (**2.27 **mm median) 3D distal tip error**, executing in 4.15 **s on a 24-core workstation CPU** (1.26 s on standard-resolution scans, 5.85 s on high-resolution scans) and 2.67 **s on a GPU**. On the 759 full-depth active-dwell shafts it reaches **91.3% recall** (94.7% at $\tau = 35\%$) and **90.0% obturator-adjusted F1** (**90.5%** with inner-OOF cross-fitting), and on the 23 patients with severe catheter adhesion it reaches **91.3%–91.8% F1**. Across all 40 doubly annotated clinical scans, Grubbs' three-cornered-hat analysis (0.39 mm intrinsic shaft MAD SD vs. 0.35 and 0.48 mm for the two human teams; 2.98 vs. 3.55 mm tip MAD σ_z), one-sided non-inferiority tests ($p_{\text{pt}} \leq 0.0085$, $p_{\text{cath}} \leq 2.5 \times 10^{-11}$), and independent Pelvic CT adjudication (52.0% of apparent extra MRI tracks confirmed on CT; 4.20 mm median tip error vs. 4.31 and 8.00 mm on disputed tips) establish that the algorithm is statistically non-inferior to human expert variability. Coupled with calibrated split-conformal 3D tip uncertainty intervals (90.2% empirical coverage at 90% nominal), reviewing 2 to 8 flagged tips per patient in simulated 1-click tip review reduces P95 tip error below 10 mm and holds counterfactual AAPM TG-43U1 ^{192}Ir target $|\Delta D_{90}|$ to 2.8% median (5.9% mean), while zero-shot transfer generalizes to 3T prostate MRI (96.3%–100.0% F1), 3D wbSSFP (92.9% F1), 4D serial insertion tracking (82.8% pooled F1, +12.4% precision gain, up to 93.3% F1 on `Case 070`), and 38 paired Pelvic CT patients (40 CT volumes total) without retraining.

DATA AVAILABILITY

The 54 clinical MRI scans and their expert traces were acquired at Brigham and Women's Hospital under Institutional Review Board approval and contain protected health information; they cannot be shared publicly. De-identified trajectory annotations and per-patient evaluation results may be made available by the corresponding author on reasonable request, subject to institutional and ethics approval. The 48 needle-free scans are the public Medical Segmentation Decathlon prostate task (Task05_Prostate), available at http://medicaldecathlon.com [81]. The SynthBrachy3T generator is released with the code, so the 300 synthetic bundles can be regenerated from a fixed random seed.

CODE AVAILABILITY

The 3D Slicer extension, the Rust Stage 0 kernel, and the exported ONNX and NumPy models are available at https://github.com/needlefinder/NeedleFinder under an open-source license. Training, nested cross-validation, statistical analysis, SynthBrachy3T, prediction-export, and figure-generation code is collected in the `vault/paper/benchmarks` directory of the same repository and will be made public on publication.

ETHICS STATEMENT

This retrospective study used imaging data acquired during routine clinical care under a protocol approved by the Institutional Review Board of Brigham and Women's Hospital / Mass General Brigham [protocol number to be inserted].

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CONFLICTS OF INTEREST

[The authors declare no conflicts of interest, or list them here.]

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